

The Bio-blunder and Policy Failure: A case study of Sri Lanka.

Bachelor Thesis

Supervisor:

Ing. Pawan Kumar Mishra, Ph.D

Gaya Shubhani Jayasinghe

Brno 2023

I would like to sincerely thank Ing. Pawan Kumar Mishra, Ph.D., my thesis supervisor, for all of his guidance and advice. I also appreciate Ankit Tripathi, an undergraduate intern in master's studies, for offering his assistance. I am grateful to Dr. Mishra, my mentor and supervisor, for his important guidance and encouragement. Their knowledge and support have been invaluable in helping me shape this thesis. I am appreciative of my friends Sanya Goswami and Emmanuel Elias, and especially my family and husband Anuradha Abhayawardhane, for their constant support and understanding throughout this challenging yet successful journey.

Statutory declaration

I hereby declare that this thesis entitled: **The Bio-blunder and Policy Failure: A case study of Sri Lanka** was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final thesis.

I am aware that my thesis is written in accordance to Act. 121/2000 Coll., on Copyright and therefore Mendel University in Brno has the right to conclude licence agreements on the utilization of the thesis as a school work in accordance with Article 60(1) of the Copyright Act.

Before concluding a licence agreement on utilization of the work by another person, I will request a written statement from the university that the licence agreement is not in contradiction to legitimate interests of the university, and I will also pay a prospective fee to cover the cost incurred in creating the work to the full amount of such costs.

In Brno on 31st December 2023

Abstract

Jayasinghe, G.S. The Bio-blunder and Policy Failure: A case study of Sri Lanka. Bachelor thesis. Brno: Mendel University, 2023.

This research delves into the complexities within Sri Lanka's bio-based industries, examining how policy frameworks, technological innovations, stakeholder engagement, and market dynamics interconnect. It sheds light on the significant consequences of bio-blunders and policy gaps, aiming to comprehend the root causes of the economic crisis impacting the agricultural sector. The study employs a robust methodology, blending conceptual frameworks, correlational analyses, and detailed surveys to unveil the multifaceted challenges these industries face.

Chapter-wise exploration reveals layers of industry intricacies, starting from dissecting its landscape to identifying the crisis symptoms. Methodological chapters detail frameworks, hypotheses, and research designs, culminating in meticulous data analysis utilizing statistical tools and comparative methods. The findings emphasize the pivotal role of various elements in driving sustainability. Technological advancements emerge as a linchpin, exerting a profound influence on industry sustainability. While policy frameworks wield a moderate impact, stakeholder engagement, and market access are pivotal for sustainable growth. The study proposes actionable recommendations directed at stakeholders and policymakers. These suggestions advocate for targeted investments in technology, strengthened collaboration among stakeholders, refined policy structures, and fostering a culture of continuous improvement. Implementing these suggestions could trigger a transformative shift toward sustainability in Sri Lanka's bio-based industries.

Furthermore, the research spotlights promising avenues for future exploration. It invites scholars and industry experts to delve deeper into policy analyses, technological assessments, stakeholder dynamics, market access strategies, employee satisfaction correlations, and interdisciplinary approaches. These areas promise to enrich discussions on sustainable development within Sri Lanka's bio-based industries. In essence, this thesis offers a comprehensive understanding of challenges, impactful relationships, and strategic recommendations crucial for steering Sri Lanka's bio-based industries toward sustainability. Its findings contribute not only to academic discourse but also offer practical insights capable of reshaping industry practices, fostering innovation, and propelling the nation toward a more sustainable future.

Keywords

Bio-blunder, bio-based, policy failure, Sri Lanka, biosecurity, stakeholder engagement, policy making, agricultural policy

Abstrakt

Jayasinghe, G.S. Bio-omyl a selhání politiky: Případová studie Srí Lanky. Bakalářská práce. Brno: Mendelova univerzita, 2023.

Tento výzkum se zabývá složitostí srílanského průmyslu založeného na biotechnologiích a zkoumá, jak se vzájemně propojují politické rámce, technologické inovace, zapojení zúčastněných stran a dynamika trhu. Osvětluje závažné důsledky biologických chyb a mezer v politice a snaží se pochopit základní příčiny hospodářské krize, která má dopad na odvětví zemědělství. Studie využívá robustní metodologii, která kombinuje koncepční rámce, korelační analýzy a podrobné průzkumy, aby odhalila mnohostranné výzvy, kterým tato odvětví čelí.

Průzkum po kapitolách odhaluje vrstvy složitostí odvětví, počínaje pitváním jeho krajiny až po identifikaci příznaků krize. Metodologické kapitoly podrobně popisují rámce, hypotézy a výzkumné designy, které vrcholí pečlivou analýzou dat s využitím statistických nástrojů a srovnávacích metod. Zjištění zdůrazňují klíčovou roli různých prvků při řízení udržitelnosti. Technologický pokrok se ukazuje jako základní pilíř, který má hluboký vliv na udržitelnost průmyslu. Zatímco politické rámce mají mírný dopad, zapojení zainteresovaných stran a přístup na trh jsou pro udržitelný růst klíčové. Studie navrhuje praktická doporučení zaměřená na zúčastněné strany a tvůrce politik. Tyto návrhy prosazují cílené investice do technologií, posílení spolupráce mezi zúčastněnými stranami, zdokonalení politických struktur a podporu kultury neustálého zlepšování. Realizace těchto návrhů by mohla vyvolat transformační posun směrem k udržitelnosti v průmyslu založeném na biotechnologiích na Srí Lance.

Výzkum navíc upozorňuje na slibné cesty pro budoucí výzkum. Zve vědce a odborníky z oboru, aby se hlouběji ponořili do analýz politik, technologických hodnocení, dynamiky zúčastněných stran, strategií přístupu na trh, korelací spokojenosti zaměstnanců a interdisciplinárních přístupů. Tyto oblasti slibují obohacení diskusí o udržitelném rozvoji v rámci srílanského průmyslu založeného na biotechnologiích. Tato práce v podstatě nabízí komplexní porozumění výzvám, dopadovým vztahům a strategickým doporučením, která jsou zásadní pro nasměrování srílanského průmyslu založeného na biotechnologiích směrem k udržitelnosti. Její zjištění přispívají nejen k akademickému diskurzu, ale také nabízejí praktické poznatky schopné přetvořit průmyslové postupy, podpořit inovace a pohánět národ směrem k udržitelnější budoucnosti.

Klíčová slova

Bio-omyl, biologický základ, selhání politiky, Srí Lanka, biologická bezpečnost, zapojení zúčastněných stran, tvorba politiky, zemědělská politika, dotace na hnojiva, účast zemědělců

Contents

1	Introduction	12
1.1	Background of the Study	12
1.1.1	Background of the Environment.....	12
1.2	Identification of the Research Problem	13
1.2.1	Symptoms of the Problem.....	13
1.2.1.1	Harvest decrement with new policy changes regarding the bio-based farming in Sri Lanka.....	13
1.2.1.2	Poor coordination and communication.....	14
1.2.1.3	Inadequate biosecurity protocols	15
1.3	Justification of the Research Problem	15
1.3.1	Imported quantity of the Raw rice were increased at 2021-2022 period in Sri Lanka	15
1.4	Problem Statements.....	16
1.5	Research Question	16
2	Objectives	17
3	Literature Review	18
4	Methodology	25
4.1	Conceptual Framework of the Study	25
4.2	Development of Hypothesis	27
4.3	Operationalization	28
4.4	Design of the Thesis.....	29
4.4.1	Design of the sampling.....	29
4.4.1.1	Population.....	29
4.4.1.2	Sample	30
4.4.1.3	Sampling	31
4.5	Methods of data collection.....	33
4.6	Method of Data Analyzing.....	34
5	Data Presentation and Analysis	36

5.1	Respond rate of data	36
5.2	Demographic analysis of the Data.....	37
5.2.1	Categories of the Demographic analysis	37
5.2.1.1	Population Structure and Age Distribution.....	37
5.2.1.2	Regional Disparities and Urban-Rural Divide	38
5.2.1.3	Gender Dimensions	39
5.3	The Nature of responsiveness level of the Sustainable Development of Bio-based Industries.....	40
5.3.1	Grand Mean, Standard deviation & Skewness of the Sustainable Development of Bio-based Industries.....	40
5.4	Hypothesis testing for validating on Sustainable Development of Bio-based Industries.....	42
5.4.1	The impact of policy frameworks and Implementation on Sustainable Development of Bio-based Industries	42
5.4.1.1	Correlation analysis of the policy frameworks and Implementation on Sustainable Development of Bio-based Industries	42
5.4.1.2	Correlation Regression analysis of the policy frameworks and Implementation on Sustainable Development of Bio-based Industries	43
5.4.1.3	ANOVA of the policy frameworks and Implementation on Sustainable Development of Bio-based Industries	44
5.4.2	The impact of Technological Advancements and Innovations on Sustainable Development of Bio-based Industries	44
5.4.2.1	Correlation analysis of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.....	44
5.4.2.2	Regression analysis of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.....	45
5.4.2.3	ANOVA of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.	46
5.4.3	The impact of Stakeholder Engagement and Participation on Sustainable Development of Bio-based Industries	47
5.4.3.1	Correlation analysis of the Stakeholder Engagement and Participation on Sustainable Development of Bio-based Industries	47
5.4.3.2	Regression analysis of the Stakeholder Engagement and Participation on Sustainable Development of Bio-based Industries	48
5.4.3.3	ANOVA of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.	49

5.4.4	The impact of Market Opportunities and Access on Sustainable Development of Bio-based Industries.....	49
5.4.4.1	Correlation analysis of the Market Opportunities and Access on Sustainable Development of Bio-based Industries	49
5.4.4.2	Regression analysis of the Market Opportunities and Access on Sustainable Development of Bio-based Industries	50
5.4.4.3	ANOVA of the Market Opportunities and Access on Sustainable Development of Bio-based Industries.....	51
5.4.5	Multiple Regression analysis	52
5.4.6	Multiple ANOVA analysis	52
5.4.7	Multiple results of the coefficients & significant values.....	52
5.4.8	Multiple results of the coefficients & significant values.....	54
	Summary of the hypothesis testing.....	54
6	Discussion	56
7	Conclusion	58
7.1	Conclusions of the study.....	58
7.1.1	Impact & relationship between Policy Frameworks and Implementation and Sustainable Development of Bio-based Industries.....	58
7.1.2	Impact & relationship between Technological Advancements and Innovations and Sustainable Development of Bio-based Industries	59
7.1.3	Impact & relationship between Stakeholder Engagement and Participation and Sustainable Development of Bio-based Industries.....	59
7.1.4	Impact & relationship between Stakeholder Engagement and Participation and Sustainable Development of Bio-based Industries.....	60
7.1.5	Summary of the Conclusions	60
7.2	Recommendations	61
7.3	Open Avenue for other research	62
8	References	63
9	List of figures	71
10	List of tables	72
	Questionnaire Form (blank)	75

1 Introduction

This chapter provides an overview of the study's background and justification, which revolves around the bio-blunder and policy failure in Sri Lanka's bio-based industries. In recent years, the agricultural sector in Sri Lanka has encountered significant difficulties, resulting in an economic crisis. (Munasinghe et al., 2017) The main goal of this study is to examine the causes of these challenges and propose solutions for stakeholders such as policymakers, bio-based industries, and local farmers. (Cairney et al., 2023) The research problem is characterized by symptoms such as a decline in rice harvest, inadequate coordination and communication among government agencies, and insufficient biosecurity protocols. (Burchfield & Poterie, 2018), These factors have contributed to an increase in the import of raw rice and have had severe economic consequences. The research question guiding this study focuses on uncovering the underlying causes of the bio-blunder and policy failure and developing recommendations to address these issues effectively. The research objectives involve investigating the root causes, analysing the economic impact, and formulating practical recommendations for the stakeholders involved. (Ministerie van Landbouw, 2023)

1.1 Background of the Study

The agricultural sector in Sri Lanka has faced significant challenges in recent years, particularly within the bio-based industries and the local farming community. The policies implemented to support and regulate these industries have encountered numerous obstacles, resulting in an economic crisis. This thesis aims to delve into the bio-blunder and policy failure in Sri Lanka, focusing on providing recommendations to stakeholders such as policy makers, bio-based industries, and local farmers within the regional context. (Rodrigo, 2022)

1.1.1 Background of the Environment

Within the broader context of bio-based industries, it is crucial to examine the role and impact of different entities involved in this sector. This research will closely analyze the operations of companies within the bio-based industries, assessing their contributions and the challenges they face. (Roshan & Pandithage, 2023) By understanding the background of these companies, including their strategies and interactions with policy frameworks, valuable insights can be gained into the complexities surrounding the bio-based industry in Sri Lanka.

1.2 Identification of the Research Problem

This study centers around the challenges confronted by the bio-based industry in Sri Lanka, specifically addressing the policy failure that has contributed to an ongoing economic crisis. The existence of various symptoms related to this problem necessitates a comprehensive investigation. (Rodrigo, 2022)

1.2.1 Symptoms of the Problem

1.2.1.1 Harvest decrement with new policy changes regarding the bio-based farming in Sri Lanka.

For this, major three districts (Ampara, Anuradhapura & Polonnaruwa) relating to the amount of harvest of rice were selected and data was analyzed. As per the data, 2021-2022 period harvest has significant decrement indeed. (*Paddy Statistics. 2021 Yala Season. Department of Census and Statistics Sri Lanka, n.d.*)

Tab. 1 Total production of rice harvest per year

Year	Ampara District (Bushels)	Polonnaruwa District (Bushels)	Anuradhapura District (Bushels)
2016 Yala / Maha	32125	36458	24365
2017 Yala / Maha	33958	37598	25101
2021 Yala / Maha	20758	21569	15236
2022 Yala / Maha	20125	20012	14987

Source: (*Paddy Statistics. 2021 Yala Season. Department of Census and Statistics Sri Lanka, n.d.; Sri Lanka, 2022; Senanayake & Premaratne, 2016*)

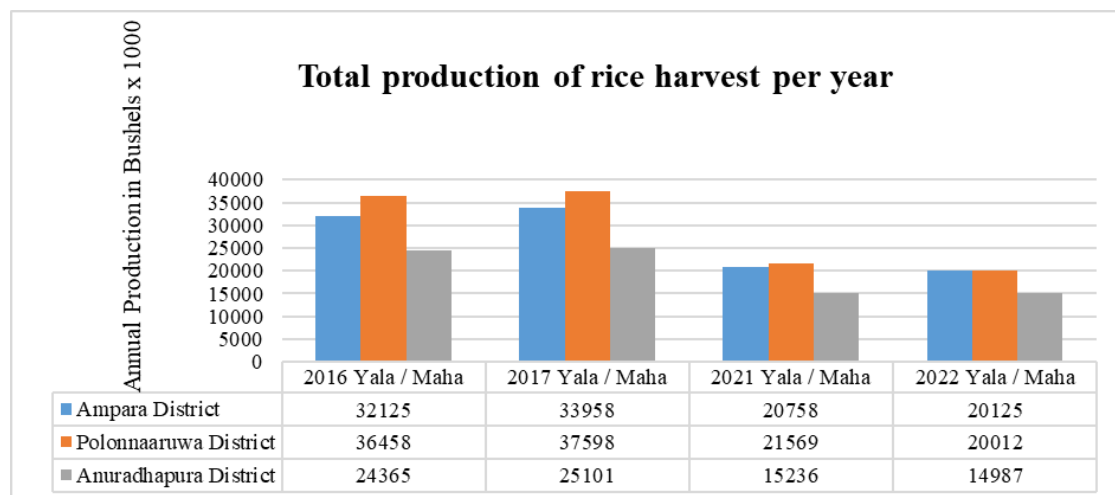


Fig. 1 Total production of rice harvest per year

Source: (*Paddy Statistics. 2021 Yala Season. Department of Census and Statistics Sri Lanka, n.d.; Sri Lanka, 2022; Senanayake & Premaratne, 2016*)

1.2.1.2 Poor coordination and communication

There was a lack of effective coordination and communication among various government agencies involved. This led to confusion and delays in implementing necessary measures to address the bio-blunder. (Wijerathna-Yapa et al., n.d.) Number of complaints regarding the bio-based farming issues were increased on 2021-2022 period after the policy changes about the bio-based farming. Timely sharing of crucial information and the development of a unified strategy were hindered as a result. (English, 2021)

Tab. 2 No. of complaints to the Ministry of agriculture regarding the bio-based farming issues per year

Year	No. of complaints
2016	172
2017	184
2018	168
2019	159
2020	189
2021	449
2022	457

Source: (Sri Lanka, 2022)

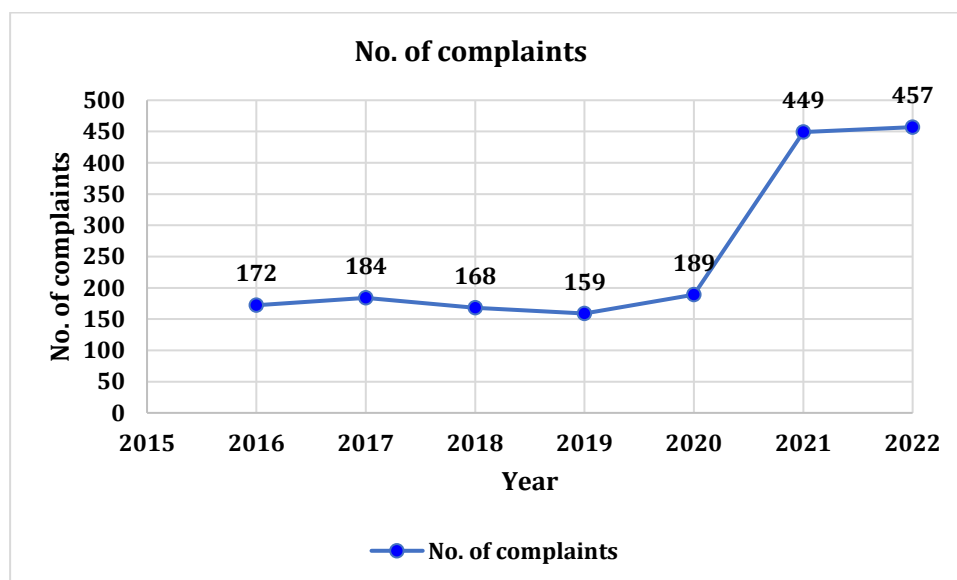


Fig. 2 No. of complaints to the Ministry of agriculture regarding the bio-based farming issues per year

Source: (Sri Lanka, 2022)

1.2.1.3 Inadequate biosecurity protocols

Sri Lanka's measures and protocols for biosafety and biosecurity were not sufficiently stringent to prevent the occurrence of the bio-blunder. (Drechsel et al., 2023) There were gaps in enforcing regulations and standards, which allowed for mishandling or accidental release of dangerous pathogens. This, in turn, led to the outbreak and subsequent spread of the problem. (Rodrigo, 2022)

1.3 Justification of the Research Problem

The research problem holds profound implications for the bio-based industries in Sri Lanka. Understanding and addressing the challenges faced by these industries are crucial for sustaining and enhancing their contribution to the national economy. (Munasinghe et al., 2017; Rathnayake et al., 2022) The following justifications further underscore the importance of investigating this problem.

1.3.1 Imported quantity of the Raw rice were increased at 2021-2022 period in Sri Lanka

As per the below mentioned data, the raw rice imported quantity increased at the 2021-2022 period relating to the other period.

Tab. 3 Raw rice production Vs imported quantity

Period	Raw rice production in Sri Lanka (million metric tons)	Imported raw rice quantity of Sri Lanka (million metric tons)
2016-2017	12.75	3.25
2021-2022	7.93	10.46

Source: (Sri Lanka, 2022)

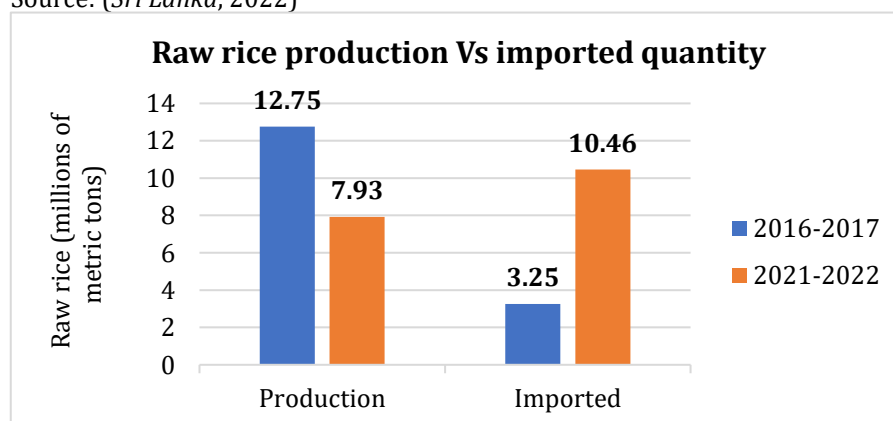


Fig. 3 Raw rice production Vs imported quantity

Source: (Sri Lanka, 2022)

1.4 Problem Statements

The bio-blunder and policy failure in Sri Lanka have had severe economic repercussions, particularly affecting the crucial bio-based industries and local farmers. Inadequate coordination, weak biosecurity protocols, and insufficient surveillance systems have impeded the timely detection and response to emerging infectious diseases. This has resulted in a delayed response and failure to effectively contain the problem, exacerbating the country's existing economic crisis. (Drechsel et al., 2023)

To address these issues comprehensively, the thesis will conduct an extensive analysis of relevant literature, utilizing bibliometric and discourse analysis methods, supported by an open-source reference manager. The Methodology chapter will outline the research process, while the Results chapter will provide insights into the magnitude of the economic crisis resulting from policy failure. The primary objective of the thesis is to develop actionable recommendations tailored to policymakers, bio-based industries, and local farmers, taking into account the regional context, with the aim of implementing effective policy interventions to mitigate the adverse impact of the bio-blunder.

1.5 Research Question

To address the identified problem, the following research question will guide this study: "What are the underlying causes of the bio-blunder and policy failure in Sri Lanka's bio-based industries, and how can recommendations be formulated to overcome these challenges?"

2 Objectives

The research objectives for this study are as follows:

To investigate the causes of the bio-blunder and policy failure in Sri Lanka's bio-based industries. Investigating the underlying reasons for the improper handling of biological sciences in Sri Lanka's bio-based sectors is part of this. It examines elements that have contributed to inefficiencies or disasters in various areas, such as insufficient implementation strategies, misaligned policies, or regulatory flaws.

To analyze the economic crisis resulting from the policy failure and its impact on the industry. This calls for a thorough analysis of the economic crisis brought on by ineffective policies in Sri Lanka's bio-based sectors. This analysis aims to comprehend the profound effects on the industry's growth, stability, investment, and general state of health.

To formulate recommendations for stakeholders, including policy makers and for bio-based industries. This stage involves integrating practical suggestions meant for various stakeholders. These recommendations will address policymakers, business executives, and other pertinent parties, providing doable actions and tactics to close policy gaps, enhance implementation processes, and promote a stronger and more sustainable bio-based sector in Sri Lanka.

3 Literature Review

In this chapter, the literature review section explores and synthesizes existing research and knowledge related to the bio-blunder and policy failure in Sri Lanka's bio-based industries. By examining scholarly articles, reports, and publications, the study aims to gather valuable insights into the underlying causes, the resulting consequences, and potential solutions for the identified issues. The literature review acts as a crucial building block for the study, providing a solid understanding of the topic and serving as a basis for developing informed recommendations.

In recent years, the intersection of bio-based industries, agriculture, and policy-making has become a crucial area of study, especially in developing countries like Sri Lanka. This literature review aims to explore existing research and scholarly discourse related to the bio-blunder and policy failure in Sri Lanka, with a focus on providing recommendations for key stakeholders, including policy makers, bio-based industries, and local farmers.

Bio-based Industries and their Impact on Agriculture

Bio-based industries, which encompass sectors such as bioenergy, biofuels, and bioplastics, have gained significant attention due to their potential to address environmental challenges and promote sustainable development. However, their implementation and impact on agriculture require careful consideration. (Gunarathne et al., 2019)

Policy Making in the Context of Bio-based Industries

Effective policy making is crucial to ensure the successful integration of bio-based industries into the agricultural sector. This section examines the key policy frameworks and challenges in Sri Lanka, shedding light on the existing policy gaps and the consequences of inadequate policy implementation. (Cairney et al., 2023; Timonina & Stephan, 2023)

The Bio-blunder: Case Study of Sri Lanka

The bio-blunder in Sri Lanka refers to the unintended consequences and negative impacts resulting from the implementation of bio-based industries and associated policies. This section presents a case study analysis of specific instances of policy failure and their implications for the agricultural sector in Sri Lanka. (Ranathilaka & Arachchi, 2019; Wijetunga & Saito, n.d.)

Stakeholder Perspectives: Policy Makers

Understanding the perspectives and roles of policy makers is essential for formulating effective recommendations. This section explores the challenges faced by policy

makers in developing and implementing bio-based industry policies, with a particular emphasis on the Sri Lankan context. (Weerahewa & Dayananda, 2023)

Stakeholder Perspectives: Bio-based Industries

The bio-based industries themselves play a critical role in the success or failure of policies. This section analyzes the perspectives of bio-based industry representatives, highlighting their experiences, challenges, and suggestions for improving the integration of these industries with agriculture in Sri Lanka. (D'Amico et al., 2022)

Stakeholder Perspectives: Local Farmers

Local farmers are among the most affected by policy failures and bio-blunders. Understanding their experiences and perspectives is essential for crafting recommendations that address their needs and concerns. (Lemke & Harris-Wai, 2015) This section explores the challenges faced by local farmers in Sri Lanka, their interactions with bio-based industries, and their suggestions for policy improvements. (Weerahewa & Dayananda, 2023)

Definition and Scope of Bio-based Industries

Bio-based industries encompass a wide range of sectors that produce goods and services using renewable biological resources. This section examines various definitions and conceptualizations of bio-based industries, clarifying the scope and distinguishing characteristics of this sector. (Upadhyay et al., 2023)

Economic and Environmental Implications of Bio-based Industries

Bio-based industries have the potential to generate economic growth, create employment opportunities, and reduce environmental impacts. This section reviews studies that analyze the economic and environmental implications of bio-based industries, including their contributions to resource efficiency, greenhouse gas emissions reduction, and sustainable development. (Escobar & Laibach, 2021)

Policy and Regulatory Frameworks for Bio-based Industries

Effective policy and regulatory frameworks are crucial for fostering the growth and development of bio-based industries. This section explores the existing policies and regulations governing bio-based industries at the national and international levels, examining their strengths, weaknesses, and implications for industry stakeholders. (Moosmann et al., 2020)

Technological Advances and Innovations in Bio-based Industries

Technological advancements play a pivotal role in driving innovation and enhancing the competitiveness of bio-based industries. This section provides an overview of

recent technological developments, such as biotechnology, biomass conversion processes, and bio-refineries, highlighting their potential to transform the industry landscape. (Asghar et al., 2022)

Market Opportunities and Challenges for Bio-based Products

The market for bio-based products is influenced by various factors, including consumer demand, market competitiveness, and policy support. This section reviews studies that examine market opportunities and challenges for bio-based products, addressing issues such as market penetration, consumer acceptance, and pricing strategies. (Dammer et al., 2013)

Sustainability Assessment of Bio-based Industries

Assessing the sustainability performance of bio-based industries is essential for guiding decision-making processes and ensuring their long-term viability. This section explores methodologies and frameworks used to evaluate the sustainability of bio-based industries, including life cycle assessment, social impact assessment, and economic feasibility analysis. (Rafiaani et al., 2018)

Challenges Faced by Local Farmers

Local farmers encounter various challenges in their daily agricultural practices, ranging from limited access to resources to climate change impacts. This section reviews studies that identify and analyze the challenges faced by local farmers, including land tenure issues, lack of access to credit and markets, and the vulnerability to climate variability. (Eriyagama et al., 2010; Esham et al., 2018)

Policy Support for Local Farmers

Effective policies and support mechanisms are essential for enhancing the resilience and livelihoods of local farmers. This section examines the policy interventions and initiatives targeting local farmers in the regional context, assessing their strengths, weaknesses, and impacts on farmer welfare and sustainability. (Wanigasundera & Atapattu, 2019)

Farmer Knowledge and Innovations

Local farmers possess valuable traditional knowledge and innovations that contribute to sustainable agriculture. This section explores studies that investigate the knowledge systems, farming practices, and innovations developed by local farmers, emphasizing the importance of recognizing and integrating their expertise into agricultural development strategies. (Abeywardana et al., 2019)

Access to Markets and Value Chains

Access to markets and participation in agricultural value chains are critical for local farmers' economic viability. This section reviews studies that examine the challenges and opportunities faced by local farmers in accessing markets, including issues related to market infrastructure, price volatility, and the role of farmer cooperatives. (Rosairo, 2023)

Sustainable Farming Practices

Local farmers often adopt sustainable farming practices that promote environmental conservation and agroecological approaches. This section explores studies that highlight sustainable farming practices employed by local farmers, including organic farming, crop diversification, and water conservation techniques. (Melvani et al., 2020)

Policy Frameworks for Bio-based Industries

Policy frameworks serve as the foundation for guiding the development and implementation of bio-based industry policies. This section examines existing policy frameworks at the national and international levels, assessing their strengths, weaknesses, and the implications for bio-based industry stakeholders. (Rathnachandra et al., 2023)

Policy Instruments and Mechanisms

Policy instruments and mechanisms are essential tools for translating policy objectives into practical actions. This section reviews studies that analyze various policy instruments and mechanisms employed in the context of bio-based industries, including subsidies, tax incentives, research and development support, and public-private partnerships. (Ferrando et al., 2022)

Stakeholder Engagement in Policy Making

Inclusive and participatory policy making processes involving relevant stakeholders are crucial for effective policy development. This section explores studies that examine the role of stakeholder engagement in bio-based industry policy making, emphasizing the importance of involving industry representatives, farmers, environmental groups, and local communities. (D'Amico et al., 2022; Pandey & Sharma, 2021)

Policy Challenges and Barriers

Policy making in the context of bio-based industries is not without challenges and barriers. This section reviews studies that identify and analyze the challenges faced by policy makers, including conflicting policy objectives, regulatory complexity, lack

of coordination among government agencies, and limited financial resources. (Amaratunga et al., 2020)

Policy Evaluation and Impact Assessment

Evaluating the effectiveness and impact of bio-based industry policies is crucial for ensuring their efficiency and guiding future policy decisions. This section explores studies that focus on policy evaluation and impact assessment methodologies, including economic evaluation, environmental assessment, and social impact analysis. (Vinci et al., 2023)

International Cooperation and Policy Learning

International cooperation and policy learning play a significant role in fostering the growth of bio-based industries. This section examines studies that explore the importance of knowledge sharing, best practices, and international collaboration in policy making, highlighting the role of international organizations and agreements in shaping bio-based industry policies. (Dietz et al., 2018)

Bio-based Industries and Policy Implementation

Bio-based industries have gained prominence in Sri Lanka as a means to promote sustainable development and address environmental challenges. This section examines studies that analyze the implementation of bio-based industry policies, highlighting instances of policy failure and their consequences for the agricultural sector. (Gunarathne et al., 2019)

Case Studies of the Bio-blunder in Sri Lanka

Specific case studies provide insights into the bio-blunder and its impacts on Sri Lanka's agriculture. This section reviews case studies that highlight instances of bio-blunders in Sri Lanka, such as the introduction of invasive species, unintended environmental degradation, and negative effects on local farmers' livelihoods. (Panneerselvam et al., 2012)

Environmental Impacts of the Bio-blunder

The bio-blunder in Sri Lanka has often resulted in unintended environmental consequences. This section explores studies that assess the environmental impacts of bio-blunders, including soil degradation, water pollution, loss of biodiversity, and the introduction of invasive species. (Gamage et al., 2023)

Socioeconomic Effects on Local Farmers

Local farmers in Sri Lanka bear the brunt of the bio-blunder's negative impacts. This section reviews studies that examine the socioeconomic effects of the bio-blunder

on local farmers, including income loss, land displacement, and challenges related to livelihood sustainability. (Williams et al., 2023)

Policy Failures and Lessons Learned

Analyzing policy failures is crucial for avoiding similar mistakes in the future. This section explores studies that identify and analyze policy failures associated with the bio-blunder in Sri Lanka, highlighting the lessons learned and recommendations for improved policy-making processes. (Wijerathna-Yapa et al., n.d.)

Community Engagement and Participation

Engaging local communities in decision-making processes can help prevent or mitigate the bio-blunder. This section examines studies that emphasize the importance of community engagement and participation in bio-based industry policy-making, empowering local stakeholders to contribute to sustainable development. (Nagothu, 2020)

Importance of Stakeholder Engagement in Bio-based Industries

Stakeholder engagement ensures that diverse perspectives are considered in the development and implementation of bio-based industry policies. This section examines studies that emphasize the importance of stakeholder engagement, highlighting its role in fostering transparency, inclusivity, and long-term sustainability. (Leipold & Petit-Boix, 2018)

Stakeholder Mapping and Analysis

Understanding the different stakeholders involved in bio-based industries and policy making is essential for effective engagement strategies. This section explores studies that discuss stakeholder mapping and analysis techniques, enabling a comprehensive understanding of stakeholder interests, power dynamics, and potential collaborations. (Dissanayake et al., 2021)

Perspectives of Bio-based Industry Stakeholders

Bio-based industry stakeholders, including policymakers, industry representatives, and environmental groups, have diverse perspectives and interests. This section reviews studies that explore the perspectives of various stakeholders, examining their priorities, challenges, and expectations regarding policy making, industry development, and the role of local farmers. (Lemke & Harris-Wai, 2015)

Local Farmers as Stakeholders in Bio-based Industries

Local farmers play a crucial role in bio-based industries and should be actively involved in decision-making processes. This section examines studies that highlight

local farmers' perspectives as stakeholders, addressing their needs, challenges, and aspirations, and emphasizing the importance of their knowledge and expertise. (Ranathilaka & Arachchi, 2019)

Collaborative Governance and Multi-stakeholder Partnerships

Collaborative governance and multi-stakeholder partnerships are effective approaches for engaging stakeholders in bio-based industry policy making. This section explores studies that discuss the benefits, challenges, and best practices of collaborative governance models, emphasizing the importance of building trust, fostering dialogue, and facilitating meaningful participation. (Kurian & Kojima, 2021)

Strengthening Stakeholder Engagement for Sustainable Development

Enhancing stakeholder engagement practices is crucial for achieving sustainable development goals in bio-based industries. This section examines studies that propose strategies and frameworks to strengthen stakeholder engagement, including capacity building, knowledge sharing, and mechanisms for continuous feedback and evaluation. (Filho & Brandli, 2016)

4 Methodology

In this chapter, the study framework and research design are discussed. The research framework is developed based on the findings and reviews presented in Chapter Four. It provides a structured and organized approach to the research, taking into account the objectives of the study. The research design is tailored to align with the specific research objectives and guides the overall methodology and approach for conducting the study. By establishing a clear framework and research design, the study aims to ensure a systematic and focused investigation of the bio-blunder and policy failure in Sri Lanka's bio-based industries.

4.1 Conceptual Framework of the Study

Based on the literature review, the conceptual framework of this study focuses on four key factors as independent variables, along with one dependent variable. The independent variables include policy frameworks and implementation, stakeholder engagement and participation, technological advancements and innovations, and market opportunities and access. These variables are considered to have an impact on the sustainable development of bio-based industries in Sri Lanka, which serves as the dependent variable.

Policy frameworks and implementation refer to the existing policies and their effective execution within the bio-based industries. Stakeholder engagement and participation involve the involvement of various stakeholders, such as policy makers, industry representatives, and local farmers, in the decision-making processes related to bio-based industries. Technological advancements and innovations consider the progress made in areas like biotechnology, biomass conversion processes, and bio-refineries within the bio-based industries. Market opportunities and access refer to the availability of markets for bio-based products and the extent to which local farmers can access these markets and value chains.

The dependent variable is the sustainable development of bio-based industries, which encompasses aspects such as economic growth, environmental conservation, and social well-being. By examining the relationships between these independent variables and the dependent variable, the study aims to provide recommendations for enhancing policy frameworks, stakeholder engagement, technological advancements, and market opportunities. Ultimately, the goal is to improve the overall sustainability of bio-based industries in Sri Lanka. (Escobar & Laibach, 2021; Moosmann et al., 2020; Rafiaani et al., 2018)

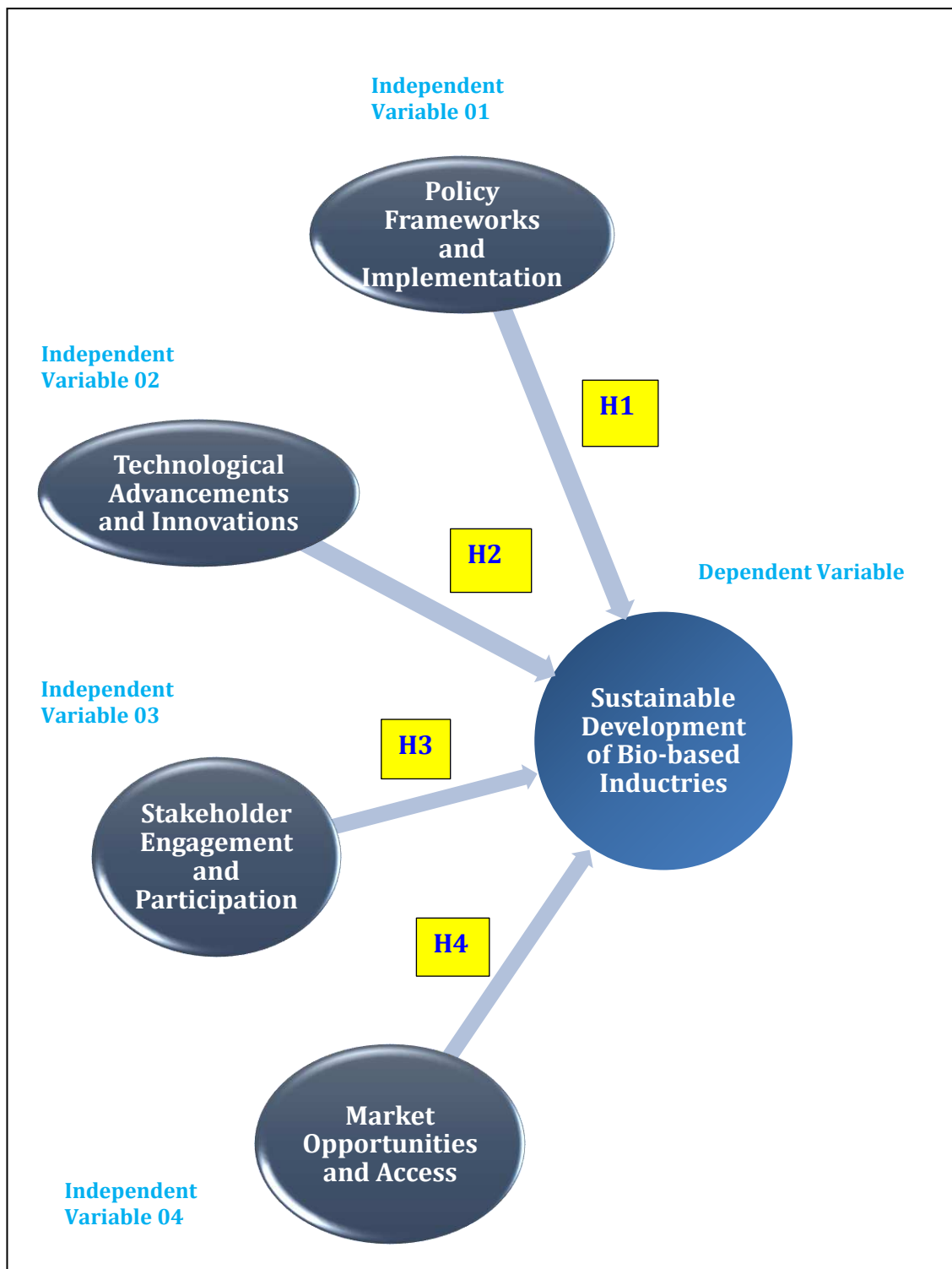


Fig. 4 Conceptual Framework of the study

4.2 Development of Hypothesis

The hypothesis was implemented as per the conceptual framework of the study.

Tab. 4 Hypothesis as per the conceptual framework of the study

H1	H1_a	There is no relationship with policy frameworks and Implementation
	H1_b	There is a relationship with policy frameworks and Implementation
H2	H2_a	There is no relationship with technological advancements and Innovations
	H2_b	There is a relationship with technological advancements and Innovations
H3	H3_a	There is no relationship with stakeholder engagement and Participation
	H3_b	There is a relationship with stakeholder engagement and Participation
H4	H4_a	There is no relationship with market opportunities and access.
	H4_b	There is a relationship with market opportunities and access.

4.3 Operationalization

The study's operationalization relies on the hypotheses mentioned earlier. It involves identifying variables, selecting measuring indicators, determining the measuring scale, and designing questionnaires.

Tab. 5 Operationalization of the study

Variable		Measuring Indicators	Measuring Scale	Questions
Independent Variable 01	Policy frameworks and implementation	Policy coherence	1 to 5 Likert scale	3.1.1,3.1.2
		Policy effectiveness	1 to 5 Likert scale	3.1.3,3.1.4
		Policy compliance and enforcement	1 to 5 Likert scale	3.1.5
Independent Variable 02	Technological advancements and innovations	Research and development investment	1 to 5 Likert scale	3.2.1,3.2.2
		Patent filings and intellectual property	1 to 5 Likert scale	3.2.3
		Technology adoption rate	1 to 5 Likert scale	3.2.4,3.2.5
Independent Variable 03	Stakeholder engagement and participation	Stakeholder representation	1 to 5 Likert scale	3.3.1
		Public awareness and engagement	1 to 5 Likert scale	3.3.2,3.3.3
		Collaboration and partnerships	1 to 5 Likert scale	3.3.4,3.3.5
Independent Variable 04	Market opportunities and access	Market growth rate	1 to 5 Likert scale	3.4.1,3.4.2
		Market competition intensity	1 to 5 Likert scale	3.4.3
		Market access barriers	1 to 5 Likert scale	3.4.4,3.4.5
Dependent Variable	Sustainable Development of Bio-based Industries	Environmental sustainability	1 to 5 Likert scale	2.1
		Economic viability	1 to 5 Likert scale	2.2,2.3
		Social impact	1 to 5 Likert scale	2.4, 2.5

4.4 Design of the Thesis

This research proposal aims to assess the effectiveness of a maintenance management system by examining its relationship with various relevant variables. The study will adopt a correlational research design, focusing on analyzing the connections between the dependent variable and independent variables. To analyze the data, a quantitative research method will be employed, utilizing formal and systematic approaches along with statistical analysis.

4.4.1 Design of the sampling

The sample designing of the research method is as follows.

4.4.1.1 Population

To determine the population of the study, which should identify the key groups or individuals who have the necessary knowledge and experience related to the bio-based industries in Sri Lanka. This includes policymakers from government agencies, representatives from bio-based industry companies, and local farmers actively involved in bio-based farming practices.

Defining the target population involves recognizing these specific groups and individuals who can provide valuable insights into the research objectives. Policymakers play a crucial role in shaping and implementing policies related to bio-based industries. Industry representatives possess firsthand knowledge of the challenges and opportunities within the sector. Local farmers have direct experience with bio-based farming practices and can provide perspectives from the grassroots level.

By identifying and including these stakeholders of the study, which can gain a comprehensive understanding of the bio-based industries in Sri Lanka and address the research objectives effectively.

N – Randomly selected 240 people from the policymakers from government agencies, representatives from bio-based industry companies, and local farmers actively involved in bio-based farming practices in Sri Lanka Therefore, N=240 people

4.4.1.2 Sample

To determine the appropriate sample size for your study, it is important to consider factors such as the desired confidence level and the variability within the population. The sample size calculator can help determine the number of participants needed to achieve a representative sample. Taking into account the research population and aiming for a 95 % confidence level, the sample size calculator takes these inputs into consideration to provide an estimate of the required sample size. By using this calculator, it is ensured that the sample is large enough to draw reliable conclusions and make meaningful generalizations from the data collected.

Therefore,

The image shows a web-based calculator titled "Determine Sample Size". It has three input fields: "Confidence Level" with radio buttons for 95% (selected) and 99%; "Confidence Interval" with a text box containing the number 5; and "Population" with a text box containing the number 240. Below these fields are two buttons: "Calculate" and "Clear". At the bottom of the form, there is a label "Sample size needed:" followed by a text box containing the number 148.

Fig. 6 Sample Calculator

Source: (Sample Size Calculator - Confidence Level, Confidence Interval, Sample Size, Population Size, Relevant Population - Creative Research Systems, n.d.)

n-148 members (According to the Sample calculator)

Calculating the sample size helps ensure the statistical validity and reliability of the findings. It allows you to gather sufficient data to analyze and draw accurate conclusions based on the characteristics of the target population. By utilizing the sample size calculator, that is determined the appropriate sample size for this study, which will enhance the credibility and robustness of the research outcomes.

$$\frac{\text{Policy makers or bio based industry or Local farmers}}{\text{Total population}} \times \text{Sample size} \quad (1)$$

4.4.1.3 Sampling

To ensure a robust and representative sample for our study, we begin by clearly defining our population, which comprises three distinct groups: policy makers, the bio-based industry, and local farmers. Understanding the proportions of each subgroup within the total population is crucial. For instance, if there are 50 policy makers, 40 members in the bio-based industry, and 150 local farmers in a total population of 240, we calculate the proportions for each group by dividing their respective counts by the overall population.

Having determined these proportions, the next step is to allocate the appropriate sample size to each subgroup. Assuming a total desired sample size of 148, we multiply the proportions by this figure to determine the number of participants for each group. This method ensures that the sample reflects the diversity of the entire population. Subsequently, within each subgroup, random sampling methods are employed to select actual participants. This randomization guarantees that our sample is representative of the unique characteristics and contexts inherent in each subgroup.

With the allocation completed and random samples drawn, the final stage involves implementing our data collection process. It's important to tailor this process to the specific attributes and nuances of each subgroup, thereby enhancing the accuracy and reliability of our study outcomes. This meticulous approach to sampling and data collection is fundamental to ensuring the statistical validity and credibility of our findings.

Example: Selected quantity of members at certain category.

$$\frac{\text{Total staff memebres of the catogery}}{\text{Total members of the study}} \times \text{Sample size} \quad (2)$$

Allocate Sample Size: Multiply each proportion by the total sample size to allocate the appropriate number of samples to each subgroup.

Total sample size of 148

- Policy makers: $(50 / 240) \times 148$
- Bio-based industry: $(40 / 240) \times 148$
- Local farmers: $(150 / 240) \times 148$

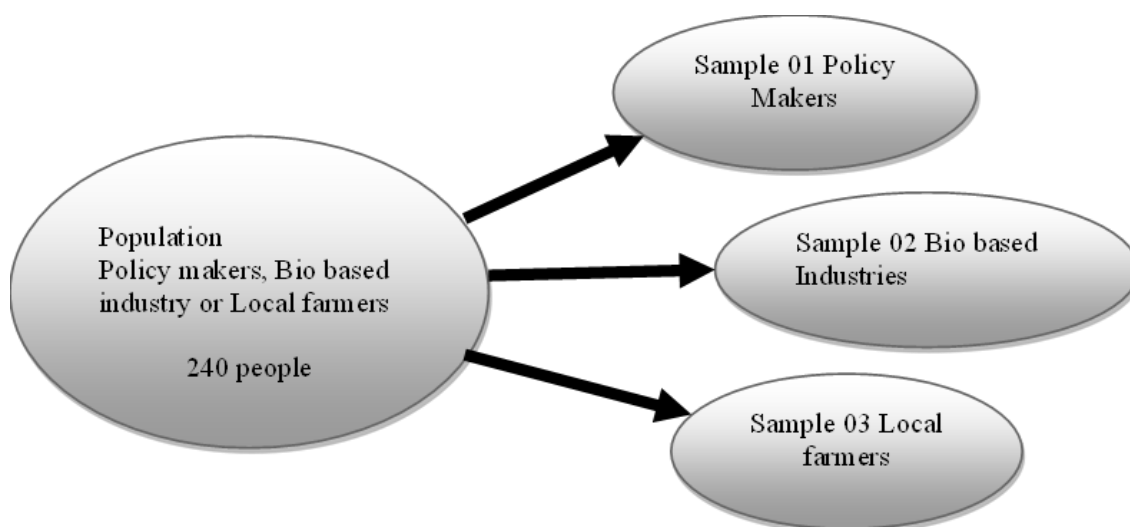


Fig. 7 Population to samples

Tab. 6 Sample calculation

Category	Policy Makers	Bio based Industries	Local Farmers
No of people in each category	50 Members	40 Members	150 Members
Selecting the number of people from each category (Stratified sampling)	$\frac{50 \times n}{N}$ $N-240$ $n-148$	$\frac{40 \times n}{N}$ $N-240$ $n-148$	$\frac{150 \times n}{N}$ $N-240$ $n-148$
Selecting people from the category (Simple Sampling)	Randomly taken from list	Randomly taken from list	Randomly taken from list
No. of people from each category (Total 148)	31 Members	25 Members	92 Members

4.5 Methods of data collection

In conducting this study, a comprehensive data collection strategy combines insights from both secondary and primary sources. Initially, the focus is on extracting valuable information from the Paddy statistics report on the Ministry of Agriculture website and the Annual report of the Ministry of Agriculture in Sri Lanka. (*Paddy Statistics. 2021 Yala Season. Department of Census and Statistics Sri Lanka, n.d.; Sri Lanka, 2022; Senanayake & Premaratne, 2016*) The primary goal of this secondary phase is to pinpoint and provide justification for the central problem under investigation. Emphasis is placed on gathering insights into influenced factors and understanding their connections with the study topic. This involves leveraging statistical data from the Ministry of Agriculture website and insights drawn from the annual report.

Subsequently, primary data collection unfolds through the utilization of a well-structured questionnaire. This primary source seeks to delve deeper into the identified problem by specifically gauging the level of commitment to the study. The questionnaire is thoughtfully crafted, aligning with the conceptual framework derived from the insights gained in the secondary phase. Careful consideration is given to ensure that the questionnaire effectively captures the significance of the identified influenced factors. Participants, representing the relevant population for the study, are then selected, and the questionnaire is administered to collect responses pertaining to the perceived importance of these factors.

The methodological approach follows a sequential progression. It begins with extracting pertinent data from secondary sources to gain a comprehensive understanding of the problem and its related influenced factors. Building upon this foundation, a questionnaire is meticulously developed to explore the level of commitment and significance of the identified factors.

Administered to the selected sample population, the questionnaire yields responses that are subsequently analyzed to uncover participants' perspectives on the influenced factors and their commitment to the study. Finally, the integration of fin-

dings from both secondary and primary data sources provides a holistic and nuanced comprehension of the identified problem, the influencing factors involved, and their intricate relationships with the study topic.

Tab. 7 Data collection summary

<i>Source of the Data</i>	<i>Objective</i>	<i>Facts</i>
<i>Secondary Data</i>		
<i>Paddy statistics report of the Ministry of agriculture website</i>	<i>To identify & justify the problem</i>	<i>Influenced factors & their relationship with the topic</i>
<i>Annual report of the Ministry of agriculture, Sri Lanka</i>	<i>To identify the problem</i>	<i>Influenced factors & their relationship with the topic</i>
<i>Primary Data</i>		
<i>Questionnaire</i>	<i>To identify the level of commitment of the study</i>	<i>Influenced factors & significance level for the study</i>

Source: (Paddy Statistics. 2021 Yala Season. Department of Census and Statistics Sri Lanka, n.d.; Sri Lanka, 2022; Senanayake & Premaratne, 2016)

4.6 Method of Data Analyzing

The data analysis for this study involved gathering primary data through a questionnaire consisting of twenty-five questions. These questions, aligned with the conceptual model and study hypotheses, were structured as multiple-choice questions with responses rated on a Likert scale ranging from 1 to 5. The Likert scale included options such as strongly disagreed (1), disagreed (2), neutral (3), agreed (4), and strongly agreed (5). The questionnaire was distributed for each category to the sample, and efforts were made during the response phase to minimize bias and obtain genuine answers from all participants.

Subsequently, the collected data were analyzed using Statistical Package for Social Science (SPSS) software (version 24) and Microsoft Excel spreadsheets. Mean

values for both independent and dependent factors were calculated, followed by the determination of grand mean, standard deviation, and coefficient of skewness. To further analyze the data, frequency distribution was presented using histograms and normality curves.

Regression analysis was employed to establish the best-fitting line for scattered diagrams, emphasizing the identification of factors contributing The Bio-blunder and Policy Failure. Descriptive statistics, including Cronbach's alpha, were then calculated to define the characteristics of the identified variables. The study prioritized regression analysis initially due to its focus on identifying factors for The Bio-blunder and Policy Failure.

The correlation between independent and dependent variables was explored using Karl Pearson correlation coefficients, indicating relationships as strongly positive, positive, negative, strongly negative, or no relationship. Separate and collective assessments of the independent factors' relationships with the dependent variable were conducted. ANOVA outputs with "F" values were then examined to assess the significance of the stated variables, with a positive "F" value suggesting a better-explained model for the study.

Finally, multiple regressions and correlations were presented to analyze the explained impact of the variables, expressed as a percentage. The remaining unexplained or residual part of the study was also highlighted in this comprehensive data analysis approach.

5 Data Presentation and Analysis

In this section, It is thoroughly examined and present the data collected from a well-crafted questionnaire. It is conducted a rigorous analysis using SPSS version 24 for statistical procedures and MS Office Excel for spreadsheet tasks. The data underwent meticulous processing in both applications to extract meaningful insights.

5.1 Respond rate of data

The response rate for the data collection was 100 %, as all 148 members of policy-makers from government agencies, representatives from bio-based industry companies, and local farmers completed the shared questionnaires. The distribution of questionnaires followed both stratified and simple sampling methods, as outlined in Chapter Four. The appendix includes the distributed questionnaire sheets and a summary of respondents.

To ensure the reliability and consistency of the collected data, the Cronbach's alpha was calculated for a selected sample of five respondents initially. If the calculated alpha exceeded the 0.6 threshold, it confirmed the consistency and reliability of the entire dataset. The results of this analysis are detailed in Table below.

Tab. 8 Results & status of the reliability test

Variable	Sample	Cronbach's Alpha	Status
Sustainable Development of Bio-based Industries	5	0.777	Consistence & Reliable
Policy frameworks and implementation	5	0.762	Consistence & Reliable
Technological advancements and innovations	5	0.750	Consistence & Reliable
Stakeholder Engagement and Participation	5	0.706	Consistence & Reliable
Market Opportunities and Access	5	0.730	Consistence & Reliable

According to the Cronbach's alpha values of the sample, the Received data set is confirmed that reliability as well as consistency.

5.2 Demographic analysis of the Data

5.2.1 Categories of the Demographic analysis

5.2.1.1 Population Structure and Age Distribution

An in-depth examination of Sri Lanka's overall population structure, encompassing an analysis of age distribution, reveals the nuanced impact of the bio-blunder and policy failure across different age groups. The repercussions were felt distinctly among the working-age population, children, and the elderly. By scrutinizing demographic trends pre – and post-events, we aim to pinpoint any discernible shifts in age demographics over time.

Tab. 9 Population Structure and Age Distribution

Age group	No of respondents	Percentage %
18 years – 25 years	27	18
25 years – 35 years	36	24
35 years – 45 years	23	16
45 years – 55 years	43	29
More than 55 years	19	13
Total	148 members	100 %

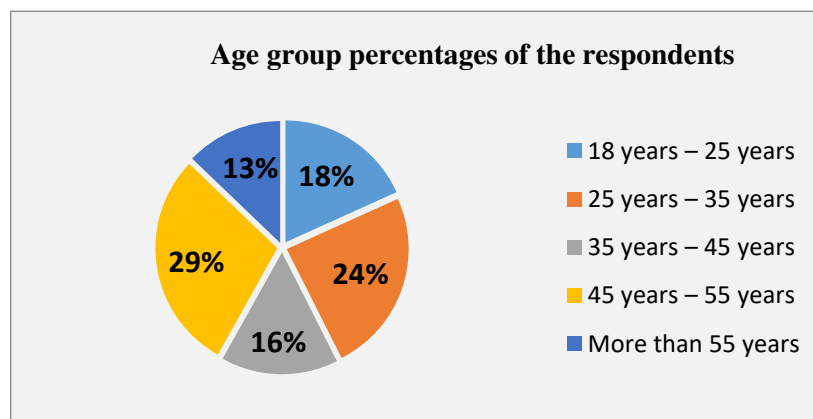


Fig. 8 Population Structure and Age Distribution

In examining the demographic composition of our survey respondents, we find a diverse distribution across various age groups. The majority of participants fall within the 45 to 55 years category, constituting 29 % of the total respondents, with a substantial count of 43 individuals. Following closely, the 25 to 35 years age group represents 24 % of the participants, totaling 36 respondents. Meanwhile, those aged

between 35 and 45 years make up 16 % of the respondents, numbering 23 individuals. The youngest cohort, comprising individuals aged 18 to 25 years, accounts for 18 % of the respondents, with 27 participants. Lastly, respondents over 55 years old constitute 13 % of the total, comprising 19 individuals. This distribution provides valuable insights into the age diversity of our surveyed population, offering a nuanced understanding of the perspectives and experiences across different life stages. The total sample size for this demographic analysis is 148 individuals.

5.2.1.2 Regional Disparities and Urban-Rural Divide

Delving into regional disparities, we explore variations in population density and demographic characteristics within Sri Lanka. The bio-blunder and policy failure manifested differently in urban and rural settings, prompting an assessment of how these disparate environments were affected. Migration patterns, influenced by the events, are analyzed, shedding light on any population movements. Disparities in healthcare access and responses are examined to uncover the differential impact on urban and rural areas.

Tab. 10 Population Structure and Age Distribution

People Region	No of respondents	Percentage %
Rural Area	86	58.1
Urban Area	24	16.2
Sub urban Area	38	25.7
Total	148 members	100 %

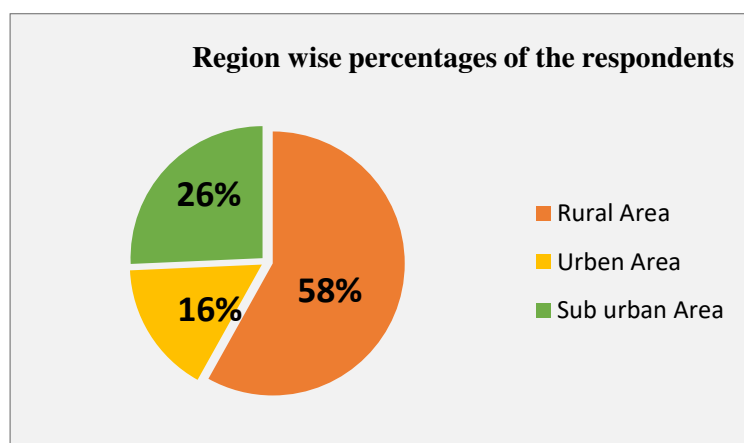


Fig. 9 Region wise percentages of the respondents

In our recent survey, we gathered insights from a diverse group of 148 individuals to understand their perspectives across different living environments. The majority of respondents, constituting 58.1 %, hailed from rural areas, reflecting a substantial representation of individuals residing in less densely populated regions. Urban areas contributed 16.2 % of the respondents, indicating the viewpoints of those living in bustling city environments. Suburban areas, with 25.7 % of respondents, played a significant role in providing insights into the experiences of individuals residing on the outskirts of urban centers. This comprehensive analysis allows us to capture a holistic view of opinions, considering the varied backgrounds of individuals from rural, urban, and suburban settings.

5.2.1.3 Gender Dimensions

An evaluation of Sri Lanka's gender distribution becomes crucial in understanding the fallout from the bio-blunder. Differential effects on men and women during the crisis are investigated to uncover nuanced gender dynamics. Changes in gender roles, particularly in healthcare and economic activities, are examined for shifts induced by the events. Special attention is given to vulnerable groups like pregnant women, as we assess the lasting consequences on gender dynamics resulting from the crisis.

These demographic analyses aim to offer a holistic comprehension of how the bio-blunder and policy failure uniquely influenced various segments of Sri Lanka's population. Beyond immediate consequences, we seek insights into the enduring implications on the country's demographic structure and dynamics.

Tab. 11 Gender Dimensions

Gender	No of respondents	Percentage %
Male	117	79
Female	31	21
Total	148 members	100 %

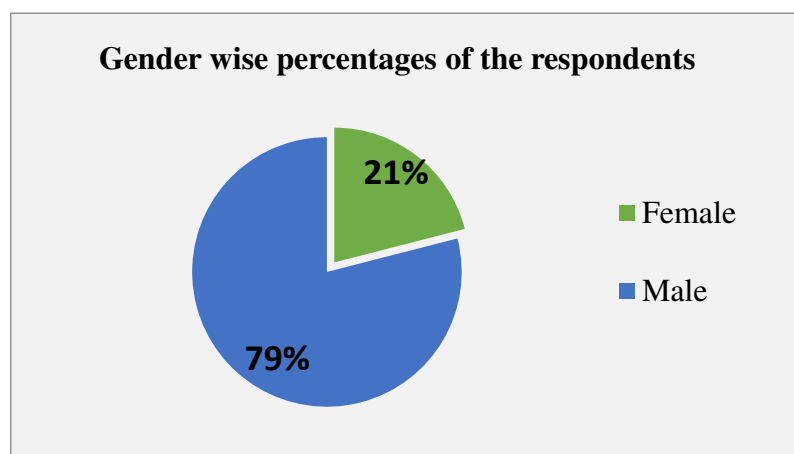


Fig. 10 Gender Dimensions

Our survey delved into the perspectives of 148 individuals across different genders, shedding light on the diversity of opinions within our sample. The majority of respondents, comprising 79 %, identified as male, while 21 % identified as female. This distribution reflects a significant representation of male participants, providing valuable insights into their experiences and viewpoints. Meanwhile, the perspectives of female respondents contribute to a well-rounded understanding of the diverse opinions present in our study. By considering the gender breakdown in our analysis, we aim to ensure a comprehensive exploration of viewpoints and experiences across genders within our surveyed population.

5.3 The Nature of responsiveness level of the Sustainable Development of Bio-based Industries

5.3.1 Grand Mean, Standard deviation & Skewness of the Sustainable Development of Bio-based Industries

The Grand mean, standard deviation & skewness of the Sustainable Development of Bio-based Industries in Sri Lanka variable were calculated due to a data sheet from the questionnaire by using the Excel.

Tab. 12 Grand Mean, Standard deviation & Skewness of the S D B I

<i>Sustainable Development of Bio-based Industries</i>	<i>Statistical data</i>
Mean	3.8310
Standard Error	0.0579
Median	4
Mode	4
Standard Deviation	0.7051
Sample Variance	0.4972
Kurtosis	0.5267
Skewness	-0.6440
Range	3.4
Minimum	1.6
Maximum	5
Sum	567
Count	148
Confidence Level (95.0 %)	0.1145

The central measure of the dependent variable in the Sustainable Development of Bio-based Industries thesis for Sri Lanka is reflected in its mean value of 3.83. This indicates that, on average, the survey responses cluster around this central point.

The standard error, a gauge of the precision of the sample mean, is relatively low at 0.058, suggesting a narrow margin of error in estimating the population mean. Both the median and mode are 4, underscoring the central tendency of the data, with half of the responses falling below this value. The standard deviation (0.705) and sample variance (0.497) indicate a moderate spread of values in the dataset, emphasizing its variability. A positive kurtosis (0.53) implies a somewhat peaked distribution, while a negative skewness (-0.64) suggests a subtle leftward asymmetry. The range, spanning from 1.6 to 5, highlights the variability within the dataset, with a maximum range of 3.4. The cumulative sum of responses is 567, gathered from a dataset comprising 148 observations. The 95 % confidence interval (3.72 to 3.94) furnishes a range wherein the true population parameter is likely to lie. In summary, these statistical observations collectively provide a thorough grasp of the distribution, central tendency, and variability of the dependent variable within the context of the Sustainable Development of Bio-based Industries in Sri Lanka.

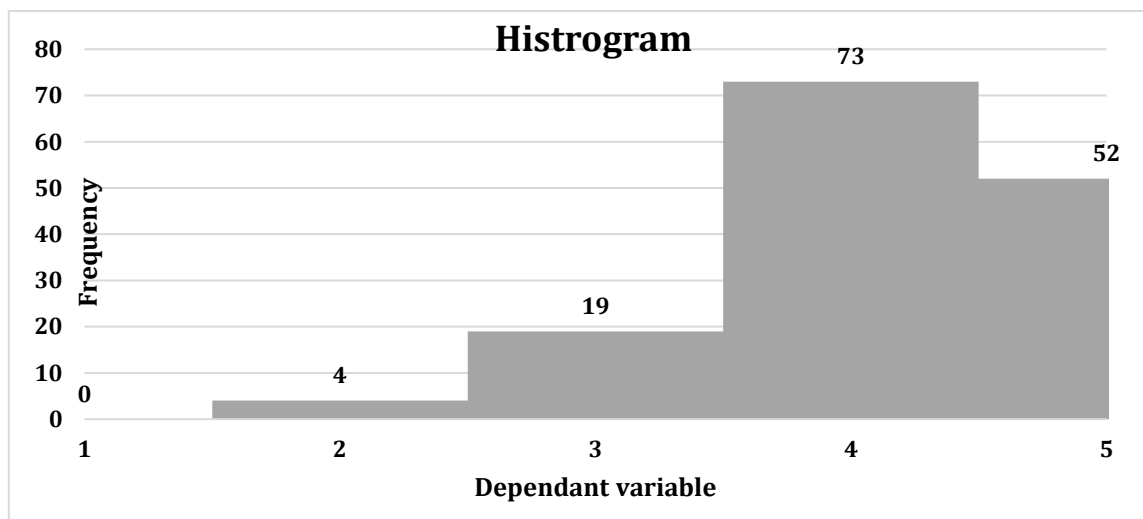


Fig. 11 Frequency distribution, Histogram & normality curve of the EMMS data

The table outlines the distribution of responses pertaining to the Sustainable Development of Bio-based Industries, employing a 1 to 5 Likert-type scale commonly utilized in surveys. This scale allows respondents to rate aspects on a spectrum from 1 (indicating the lowest or least favorable response) to 5 (representing the highest or most favorable response). The "Scale" column enumerates the different scale values, ranging from 1 to 5. The corresponding "Frequency" column quantifies how frequently each scale value was chosen in the survey or study.

For instance, a frequency of 0 for Scale 1 implies that no respondents rated the Sustainable Development variable as 1. On the other hand, Scale 2 has a frequency of 4, indicating that four respondents assessed it as a 2. The frequencies progressively increase, reaching 19 for Scale 3, 73 for Scale 4, and 52 for Scale 5. Notably, there is a category labeled "More" with a frequency of 0, the significance of

which remains unclear without additional context. It could potentially serve as a placeholder for responses falling outside the 1 to 5 scale or represent an undefined category.

This tabulated data is typically transformed into a histogram, a visual representation providing a rapid overview of the respondent distribution across the scale. A corresponding bar chart would display the scale values on the x-axis and their respective frequencies on the y-axis, facilitating a visual interpretation of the distribution of opinions regarding the Sustainable Development of Bio-based Industries.

5.4 Hypothesis testing for validating on Sustainable Development of Bio-based Industries.

5.4.1 The impact of policy frameworks and Implementation on Sustainable Development of Bio-based Industries

In section 5.4, the testing process involved a thorough analysis using various statistical methods, including regression analysis, correlation analysis, ANOVA (Analysis of Variance), and descriptive statistics. This comprehensive approach allowed us to assess and validate the hypotheses individually, providing a robust and detailed examination of the research questions.

H1_a - There is no relationship with policy frameworks and Implementation.

H1_b - There is a relationship with policy frameworks and Implementation.

5.4.1.1 Correlation analysis of the policy frameworks and Implementation on Sustainable Development of Bio-based Industries

Tab. 13 Correlation analysis of the policy frameworks and Implementation on SDBI

	<i>Sustainable Development of Bio-based Industries</i>	<i>Policy Frameworks and Implementation</i>
Sustainable Development of Bio-based Industries	1	
Policy Frameworks and Implementation	0.306911349	1

The Pearson correlation coefficient between Policy Frameworks and Implementation and Sustainable Development of Bio-based Industries is 0.307, equivalent to 30.7 %. This falls within the range of 0 to +0.5, indicating a weak positive relationship between the two variables. The significance level, denoted by the Significance

value ($P = 0.0001$), is less than 0.05, suggesting statistical significance. Consequently, the null hypothesis $H1_a$ can be rejected, and the alternative hypothesis $H1_b$ is accepted at a 95 % confidence level. This signifies that there is a statistically significant, albeit weak, positive relationship between Policy Frameworks and Implementation and Sustainable Development of Bio-based Industries.

5.4.1.2 Correlation Regression analysis of the policy frameworks and Implementation on Sustainable Development of Bio-based Industries

Tab. 14 Regression analysis of the policy frameworks and Implementation on SDBI

<i>P-value</i>	<i>Lower 95 %</i>	<i>Upper 95 %</i>	<i>Lower 95.0 %</i>	<i>Upper 95.0 %</i>
6.55608E-18	2.208671058	3.315128	2.208671	3.315128
0.000148202	0.140907016	0.430972	0.140907	0.430972

The results indicate a regression analysis assessing the influence of Policy Frameworks and Implementation on Sustainable Development of Bio-based Industries. The P-values, notably small (6.55608E-18 and 0.000148202), signify strong evidence against the null hypothesis, highlighting a statistically significant relationship between Policy Frameworks and Implementation and Sustainable Development of Bio-based Industries.

The coefficients, with their associated confidence intervals, further illuminate this relationship. For the coefficient related to Policy Frameworks and Implementation (ranging from 2.208671058 to 3.315128), and another coefficient (ranging from 0.140907016 to 0.430972), both are statistically significant.

In simpler terms, this suggests that Policy Frameworks and Implementation exert a significant impact on Sustainable Development of Bio-based Industries. The narrow confidence intervals and low P-values enhance the credibility of these findings, indicating that these results are unlikely to be attributed to random chance.

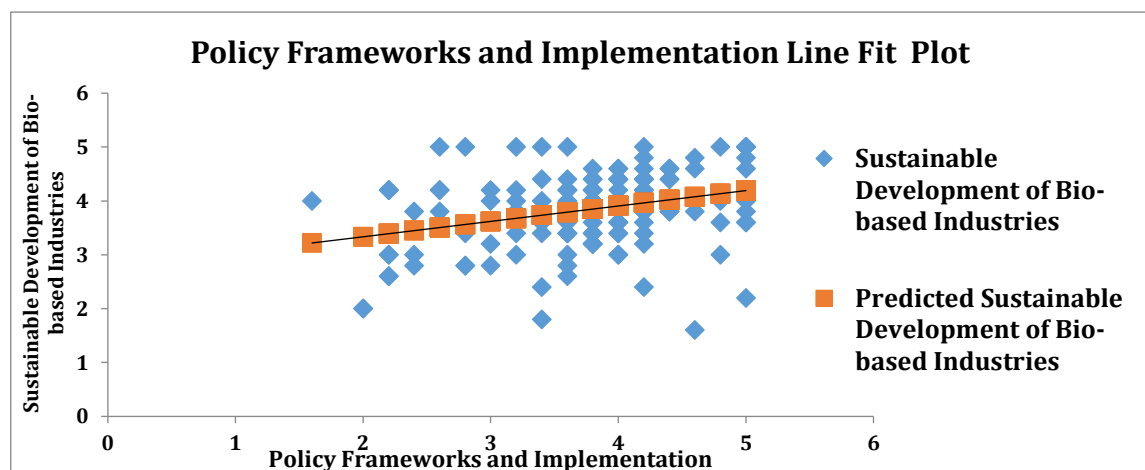


Fig. 12 Regression analysis of the policy frameworks and Implementation on SDBI

5.4.1.3 ANOVA of the policy frameworks and Implementation on Sustainable Development of Bio-based Industries

Tab. 15 ANOVA of the policy frameworks and Implementation on SDBI

<i>Model</i>	<i>Source of Variability</i>	<i>Sum of Squares</i>	<i>DF</i>	<i>Mean Squares</i>	<i>F</i>	<i>Significance F</i>	<i>P-value</i>
1	Regression	6.8853	1	6.8853	15.1825	0.0001	0.0001
	Residual	66.2117	146	0.4535	x	x	x
	Total	73.0970	147	0.4973	x	x	x

According to the ANOVA, the selected variable of policy frameworks is better with positive of “F” (F=15.183) value.

At the 5% level, the result (0.0001) is regarded as statistically significant if the P-value for the F-value is less than 0.05.

5.4.2 The impact of Technological Advancements and Innovations on Sustainable Development of Bio-based Industries

H2_a – There is no relationship with technological advancements and Innovations

H2_b – There is a relationship with technological advancements and Innovations

5.4.2.1 Correlation analysis of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries

Tab. 16 Correlation analysis of the Technological Advancements and Innovations on SDBI

	<i>Sustainable Development of Bio-based Industries</i>	<i>Technological Advancements and Innovations</i>
Sustainable Development of Bio-based Industries	1	
Technological Advancements and Innovations	0.837565624	1

Based on the analysis conducted, the Pearson correlation between Technological Advancements and Innovations and Sustainable Development of Bio-based Industries is 0.838, equivalent to 83.8 %. This places the correlation in the range of +0.8 to +1, indicating a highly positive relationship between the two variables. The significance level, represented by the Sig value (P = 0.0001), is less than the commonly accepted threshold of 0.05. Consequently, this suggests strong statistical significance in the relationship.

As a result, the null hypothesis H2_a, which likely stated no significant relationship, can be rejected. In contrast, the alternative hypothesis H2_b, which likely proposed a positive relationship, can be accepted with a 95 % confidence level. This indicates a high level of confidence that there is indeed a substantially positive association between Technological Advancements and Innovations and Sustainable Development of Bio-based Industries.

5.4.2.2 Regression analysis of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.

Tab. 17 Regression analysis of the Technological Advancements and Innovations on SDBI

<i>Regression Statistics</i>	
Multiple R	0.837566
R Square	0.701516
Adjusted R Square	0.699472
Standard Error	0.386575
Observations	148

The regression analysis conducted on the relationship between Technological Advancements and Innovations and Sustainable Development of Bio-based Industries yielded insightful results. The Multiple R, a measure of the linear association between the variables, is 0.837566, indicating a strong positive correlation. The R Square, which represents the proportion of variability in Sustainable Development of Bio-based Industries explained by Technological Advancements and Innovations, is 0.701516, suggesting that approximately 70.2 % of the variability in sustainable development can be attributed to technological advancements. The Adjusted R Square, accounting for the number of predictors in the model, is 0.699472.

The Standard Error, a measure of the accuracy of the regression predictions, is 0.386575, providing an estimate of the typical difference between the observed and predicted values. Finally, the analysis is based on 148 observations, contributing to the robustness of the findings. In essence, these results collectively indicate a substantial and statistically significant relationship between Technological Advancements and Innovations and Sustainable Development of Bio-based Industries, emphasizing the importance of technological progress in influencing sustainable outcomes in this context.

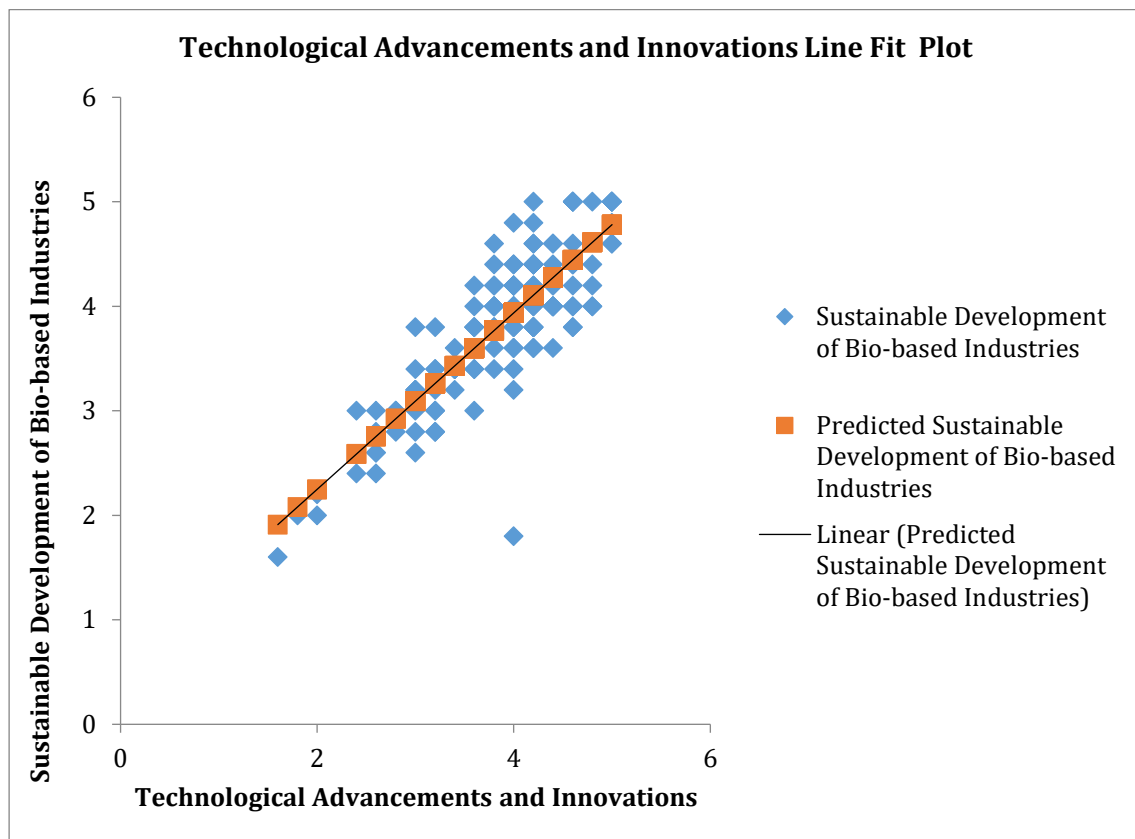


Fig. 13 Regression analysis of the Technological Advancements and Innovations on SDBI

5.4.2.3 ANOVA of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.

Tab. 18 ANOVA of the Technological Advancements and Innovations on SDBI

Model	Source of Variability	Sum of Squares	DF	Mean Squares	F	Significance F	P-value
2	Regression	51.2787468	1	51.2787	343.1387	3.66421E-40	3.66421E-40
	Residual	21.81828023	146	0.1494	x	x	x
	Total	73.09702703	147	0.4973	x	x	x

According to the ANOVA, the selected variable of policy frameworks is better with positive of “F” ($F=343.14$) value.

At the 5% level, the result ($3.66421E-40$) is regarded as statistically significant if the P-value for the F-value is less than 0.05.

5.4.3 The impact of Stakeholder Engagement and Participation on Sustainable Development of Bio-based Industries

H3_a – There is no relationship with Stakeholder Engagement and Participation

H3_b – There is a relationship with Stakeholder Engagement and Participation

5.4.3.1 Correlation analysis of the Stakeholder Engagement and Participation on Sustainable Development of Bio-based Industries

Tab. 19 Correlation analysis of the Stakeholder Engagement and Participation on SDBI

	<i>Sustainable Development of Bio-based Industries</i>	<i>Stakeholder Engagement and Participation</i>
Sustainable Development of Bio-based Industries	1	
Stakeholder Engagement and Participation	0.695430072	1

The Pearson correlation coefficient of 0.695 between Stakeholder Engagement and Participation and the Sustainable Development of Bio-based Industries implies a moderate positive relationship between these variables. As this value falls between +0.6 and +0.8, it characterizes a moderate level of positive correlation. Additionally, the significance value of 0.0001, which is lower than the commonly used threshold of 0.05, indicates a high level of confidence in the relationship observed. Consequently, based on this statistical analysis at a 95 % confidence level, the null hypothesis H3_a, which suggests no relationship between Stakeholder Engagement and Participation and the Sustainable Development of Bio-based Industries, can be rejected.

Meanwhile, the alternative hypothesis H3_b, proposing a relationship between these variables, can be accepted, supported by the statistical evidence indicating a significant and positive association between stakeholder engagement and the sustainable development of bio-based industries.

5.4.3.2 Regression analysis of the Stakeholder Engagement and Participation on Sustainable Development of Bio-based Industries

Tab. 20 Regression analysis of the Stakeholder Engagement and Participation on SDBI

<i>Regression Statistics</i>	
Multiple R	0.69543
R Square	0.483623
Adjusted R Square	0.480086
Standard Error	0.50846
Observations	148

This analysis provides crucial insights into the relationship between Stakeholder Engagement and Participation and Sustainable Development of Bio-based Industries. The multiple R-value of 0.695 signifies a strong positive correlation between these variables, essentially indicating that as stakeholder engagement increases, so does the sustainable development of bio-based industries.

The R-squared value of 0.484, or 48.4 %, tells us that almost half of the variations in the sustainable development of bio-based industries can be attributed to changes in stakeholder engagement and participation. This highlights the significant impact that stakeholder involvement has on driving sustainability in bio-based industries. The adjusted R-squared value, slightly lower at 0.480, considers the number of predictors in the model but still underscores the substantial relationship between stakeholder engagement and sustainable development in bio-based industries.

The standard error, around 0.508, indicates the average difference between the predicted and observed outcomes. A lower standard error suggests that the model's predictions are closer to the actual outcomes, enhancing the reliability of the analysis. Collectively, these statistics strongly support the notion that active engagement and participation of stakeholders significantly influence and contribute to the sustainable development of bio-based industries.

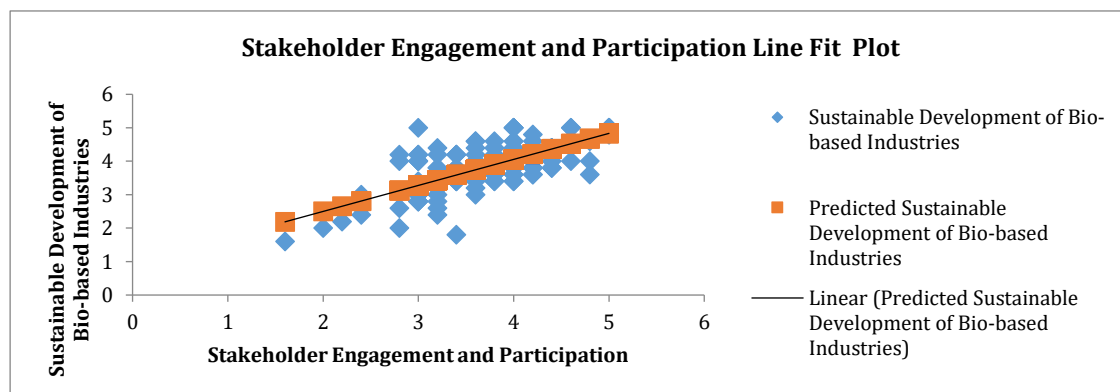


Fig. 14 Regression analysis of the Stakeholder Engagement and participation on SDBI

5.4.3.3 ANOVA of the Technological Advancements and Innovations on Sustainable Development of Bio-based Industries.

Tab. 21 ANOVA of the Stakeholder Engagement and Participation on SDBI

<i>Model</i>	<i>Source of Variability</i>	<i>Sum of Squares</i>	<i>DF</i>	<i>Mean Squares</i>	<i>F</i>	<i>Significance F</i>	<i>P-value</i>
3	Regression	35.35140243	1	35.3514	136.7392	1.04778E-22	1.04778E-22
	Residual	37.7456246	146	0.2585	x	x	x
	Total	73.09702703	147	0.4973	x	x	x

According to the ANOVA, the selected variable of policy frameworks is better with positive of “F” (F=136.74) value.

At the 5% level, the result (1.04778E-22) is regarded as statistically significant if the P-value for the F-value is less than 0.05.

5.4.4 The impact of Market Opportunities and Access on Sustainable Development of Bio-based Industries

H4_a – There is no relationship with Market Opportunities and Access

H4_b – There is a relationship with Market Opportunities and Access

5.4.4.1 Correlation analysis of the Market Opportunities and Access on Sustainable Development of Bio-based Industries

Tab. 22 Correlation analysis of the Market Opportunities and Access on SDBI

	<i>Sustainable Development of Bio-based Industries</i>	<i>Market Opportunities and Access</i>
Sustainable Development of Bio-based Industries	1	
Market Opportunities and Access	0.504907082	1

The Pearson correlation coefficient of 0.505 between Market Opportunities and Access and Sustainable Development of Bio-based Industries indicates a moderately positive relationship.

This means that as market opportunities and access increase, there tends to be a corresponding positive impact on the sustainable development of bio-based industries.

The significance value of 0.0001 being lower than the commonly used threshold of 0.05 signifies a high level of confidence in this relationship. Consequently, at a 95 % confidence level, the null hypothesis H4_a, suggesting no relationship between Market Opportunities and Access and Sustainable Development of Bio-based Industries, can be rejected.

Instead, the alternative hypothesis H4_b, proposing a relationship between these variables, can be accepted. This statistical evidence supports the idea that changes in market opportunities and access are associated with notable impacts on the sustainable development of bio-based industries, validating the proposed relationship between these factors.

5.4.4.2 Regression analysis of the Market Opportunities and Access on Sustainable Development of Bio-based Industries

Tab. 23 Regression analysis of the Market Opportunities and Access on SDBI

<i>Regression Statistics</i>	
Multiple R	0.504907
R Square	0.254931
Adjusted R Square	0.249828
Standard Error	0.610761
Observations	148

This analysis dives into the connection between Stakeholder Engagement and Participation and Sustainable Development of Bio-based Industries, uncovering important statistical insights. The multiple R-values of 0.505 highlights a moderate positive correlation between stakeholder engagement and the development of bio-based industries sustainably. The R-squared value, at 25.5 %, indicates that about a quarter of the changes seen in sustainable development outcomes within bio-based industries can be attributed to variations in stakeholder engagement and participation. This underlines the substantial impact that active involvement of stakeholders has on shaping sustainable practices in these industries.

The adjusted R-squared value, slightly lower at 0.250, maintains the notion that stakeholder engagement significantly contributes to understanding sustainable development outcomes within bio-based industries, considering the complexity of predictors in the model.

The standard error of around 0.611 signifies the average difference between predicted and observed sustainable development outcomes. A lower standard error generally indicates a higher accuracy in predicting these outcomes. Collectively, these statistics suggest that stakeholder engagement and participation indeed hold a substantial role in influencing and contributing to the sustainable growth of bio-based industries.

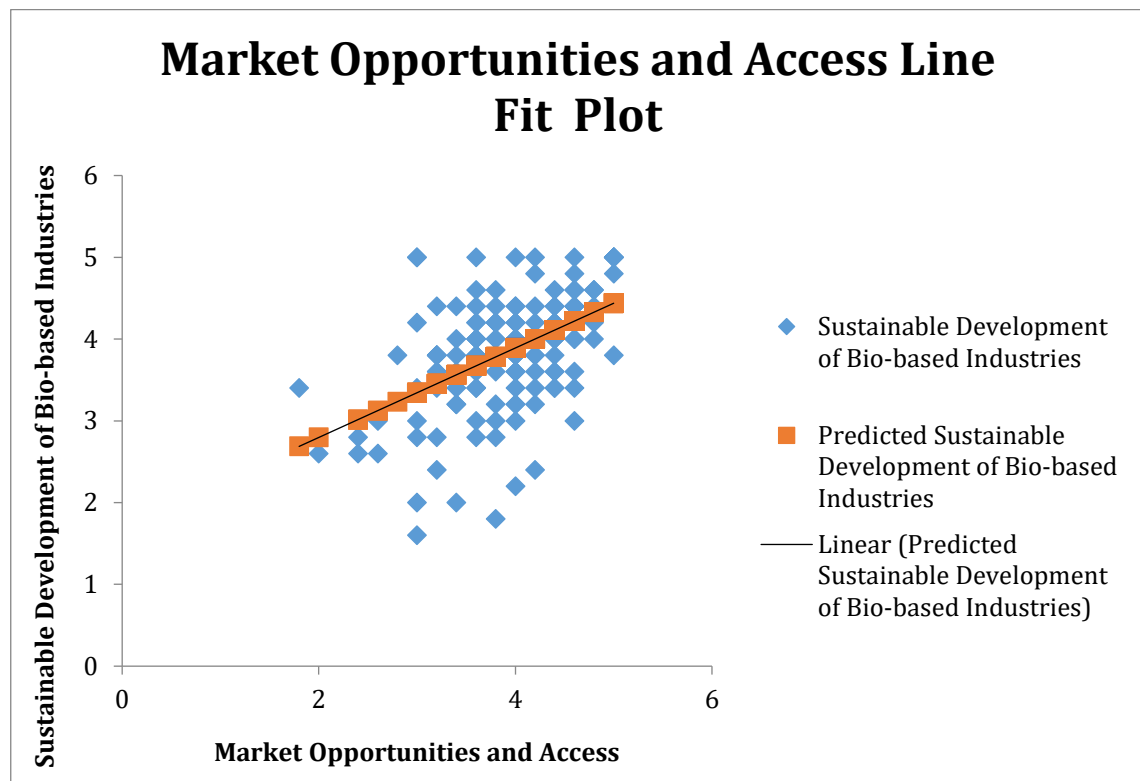


Fig. 15 Regression analysis of the Market Opportunities and Access on SDBI

5.4.4.3 ANOVA of the Market Opportunities and Access on Sustainable Development of Bio-based Industries

Tab. 24 ANOVA of the Market Opportunities and Access on SDBI

<i>Model</i>	<i>Source of Variability</i>	<i>Sum of Squares</i>	<i>DF</i>	<i>Mean Squares</i>	<i>F</i>	<i>Significance F</i>	<i>P-value</i>
4	Regression	18.63470997	1	18.6347	49.9550	5.99724E-11	5.99724E-11
	Residual	54.46231706	146	0.3730	x	x	x
	Total	73.09702703	147	0.4973	x	x	x

According to the ANOVA, the selected variable of policy frameworks is better with positive of “F” (F=49.955) value.

At the 5% level, the result (5.99724E-11) is regarded as statistically significant if the P-value for the F-value is less than 0.05.

5.4.5 Multiple Regression analysis

Tab. 25 Multiple Regression analysis

<i>Regression Statistics</i>	
Multiple R	0.851157722
R Square	0.724469467
Adjusted R Square	0.716762319
Standard Error	0.375289751
Observations	148

5.4.6 Multiple ANOVA analysis

Tab. 26 Multiple ANOVA analysis

<i>Model</i>	<i>Source of Variability</i>	<i>Sum of Squares</i>	<i>DF</i>	<i>Mean Squares</i>	<i>F</i>	<i>Significance F</i>	<i>P-value</i>
4	Regression	52.957	4	13.239	94.0000	4.95E-39	4.95E-39
	Residual	20.14	143	0.1410	x	x	x
	Total	73.097	147	0.4973	x	x	x

According to the ANOVA, the selected variable of policy frameworks is better with positive of “F” (F= 94.0000) value.

At the 5% level, the result (4.95E-39) is regarded as statistically significant if the P-value for the F-value is less than 0.05.

5.4.7 Multiple results of the coefficients & significant values

Tab. 27 Multiple results of the coefficients & significant values

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	0.268	0.2206	1.214	0.227
Policy Frameworks and Implementation	-0.055	0.0494	-1.118	0.265
Technological Advancements and Innovations	0.680	0.0667	10.191	0.000
Stakeholder Engagement and Participation	0.235	0.0737	3.188	0.002
Market Opportunities and Access	0.067	0.0621	1.085	0.280

Therefore, multiple regression model is defined as,

$Y = Constant + B_1X_1+ B_2X_2 + B_3X_3+ B_4X_4$

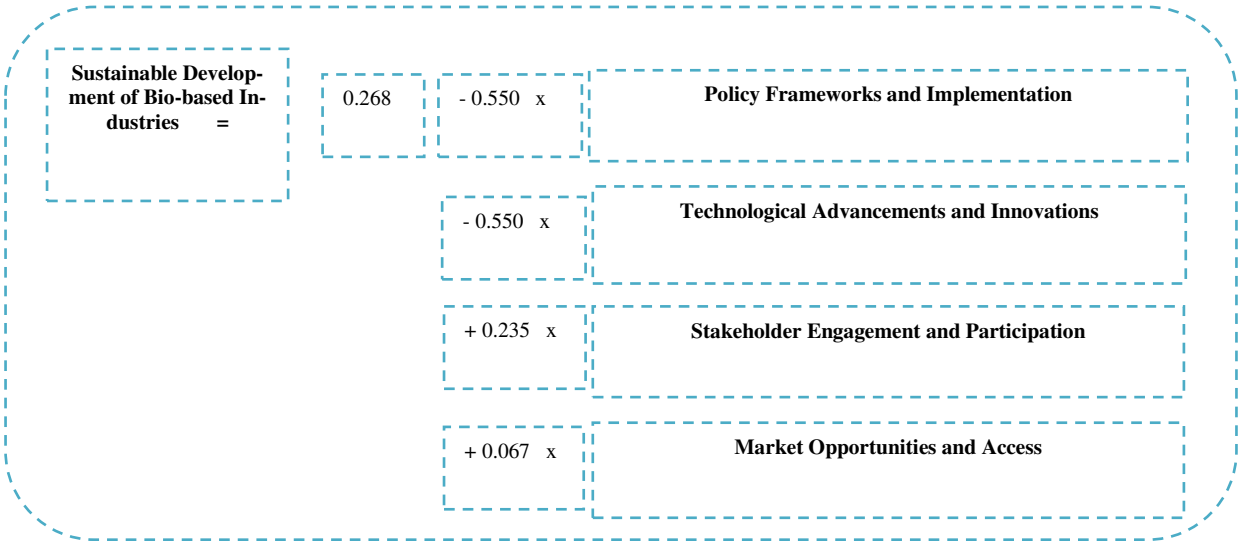


Fig. 16 Multiple Regression analysis

The R-squared value of 0.724, which translates to 72.4 %, suggests that the independent variables considered in the study collectively explain 72.4 % of the impact on Sustainable Development in Bio-based Industries. This means that around 27.6 % of the impact is influenced by other factors not included in this analysis.

The higher positive "F" value indicates a stronger suitability and effectiveness in selecting the variables for this study.

5.4.8 Multiple results of the coefficients & significant values

Summary of the hypothesis testing

Tab. 28 Summary of the hypothesis test, Source: SPSS

Variable	Significance P Value	Pearson correlation	Relationship	R Squared Value	Chance to impact on Sustainable Development of Bio-based Industries	F value
Policy Frameworks and Implementation	0.000	0.307	Weekly Positive	0.094	9.4%	15.18
Technological Advancements and Innovations	0.000	0.838	Strongly Positive	0.702	70.2%	343.1
Stakeholder Engagement and Participation	0.000	0.695	Moderately Positive	0.484	48.4%	136.7
Market Opportunities and Access	0.000	0.505	Moderately Positive	0.255	25.5%	49.95
Multiple analysis	0.000	0.851	Strongly Positive	0.724	72.4%	94

This table presents a comprehensive breakdown of the study's independent variables and their connection to the Sustainable Development of Bio-based Industries. Each variable's significance is gauged by its p-value, where a lower value, like 0.000, indicates a robust statistical significance, indicating that the observed relationships are unlikely to be random chance.

The Pearson Correlation coefficient is another crucial measure, revealing the strength and direction of the linear relationship between each independent variable and the Sustainable Development of Bio-based Industries. The Relationship column further elaborates on this connection, categorizing it as Weekly Positive, Strongly Positive, or Moderately Positive.

Furthermore, the R Squared value is instrumental in understanding how much of the variation in sustainable development can be attributed to each independent variable. For instance, a 70.2 % R-squared value for Technological Advancements and Innovations suggests that 70.2 % of the variability in sustainable development is influenced by this variable.

The Chance to Impact on Sustainable Development section delves deeper, indicating the percentage of impact on Sustainable Development explained by each variable. For instance, if Technological Advancements and Innovations account for 70.2 % of the impact, Policy Frameworks and Implementation contribute 9.4 %.

Lastly, the F value showcases the overall model fit, where a higher value, such as 343.1 for Technological Advancements and Innovations, signifies a better alignment of the model with the data. In essence, this table provides a detailed insight into the statistical significance, strength, and impact of various independent variables on the Sustainable Development of Bio-based Industries.

6 Discussion

Chapter One serves as an introduction to the study, which examines the issues of bio-blunder and policy failure within Sri Lanka's bio-based industries. Over the past years, the agricultural sector in Sri Lanka has encountered significant difficulties, resulting in an economic crisis. This study aims to understand the causes of these challenges and propose effective solutions for stakeholders such as policymakers, bio-based industries, and local farmers.

The research problem revolves around several symptoms that highlight the existing issues. These symptoms include a decline in rice harvest, inadequate coordination and communication among government agencies, and insufficient biosecurity protocols. As a result, there has been a surge in the import of raw rice, leading to severe economic consequences. The central research question guiding this study seeks to uncover the underlying causes of the bio-blunder and policy failure while developing practical recommendations to address these issues effectively (Singh, 2022; Vinci et al., 2023).

The research objectives encompass investigating the root causes of the problem, analyzing the economic impact it has caused, and formulating actionable recommendations for the stakeholders involved.

Chapter One also delves into the background of the study, specifically focusing on Sri Lanka's bio-based industries. The operations of companies within this sector are examined to gain insights into their contributions and the challenges they face. It is essential to understand the context of these companies, including their strategies and interactions with policy frameworks, to gain a comprehensive understanding of the complexities surrounding Sri Lanka's bio-based industry.

The research problem is identified as the policy failure and bio-blunder within Sri Lanka's bio-based industries, which has resulted in an ongoing economic crisis. The symptoms of the problem, including the decrement in rice harvest, poor coordination and communication among government agencies, and inadequate biosecurity protocols, further emphasize the need for a comprehensive investigation. The justification of the research problem lies in the increased import of raw rice and its impact on the bio-based industries. The problem statement highlights the severe economic repercussions caused by the bio-blunder and policy failure.

Chapter Four focuses on the research methodology, outlining the conceptual framework and research design. The conceptual framework identifies key factors that play a role in the sustainable development of bio-based industries, including policy frameworks and implementation, stakeholder engagement and participation, technological advancements and innovations, and market opportunities and access. These factors are considered independent variables that impact the dependent variable, which is the sustainable development of bio-based industries. (Fobbe & Hilltoft, 2021; Wanigasundera & Atapattu, 2019)

Hypotheses are developed based on the conceptual framework to test the relationships between the variables. The operationalization of the study involves identifying variables, selecting appropriate measuring indicators, determining the measuring scale, and designing a questionnaire to collect relevant data.

The research design adopts a correlational research design to analyze the relationships between the dependent and independent variables. The sample size is determined using a sample size calculator, and participants are selected from the target population. Data collection will be conducted through questionnaires using Likert scale responses (Råheim et al., 2016).

Within the fourth chapter, we create a platform for an extensive and thoughtful dialogue centered around our research findings. This segment serves as a space where we meticulously dissect and scrutinize the outcomes we've unearthed, employing a metaphorical magnifying glass to delve into their intricacies. As we navigate this chapter, we weave together a comprehensive perspective by juxtaposing our conclusions with data sourced from the European Union and other relevant origins. This amalgamation of information results in a refined and nuanced comparative analysis.

Data analysis will involve utilizing statistical software such as SPSS and Excel to analyze the collected data. Statistical measures like the Karl Pearson correlation coefficient will be employed to assess the relationships between variables. Regression analysis will be conducted to create predictive models, and characteristics analysis and descriptive statistics will provide further insights.

In summary, the research methodology focuses on the conceptual framework, hypotheses, operationalization, sample design, data collection method, and data analysis. By employing these elements, the study aims to conduct a systematic and focused investigation into the bio-blunder and policy failure in Sri Lanka's bio-based industries.

7 Conclusion

The concluding chapter of this study encapsulates the key discoveries gleaned from the preceding chapters. It specifically focuses on analyzing the relationship between Stakeholder Engagement and Participation and the sustainable development of bio-based industries. Drawing from these findings, the chapter extrapolates conclusions, offering practical guidance aimed at stakeholders and policymakers. These recommendations aim to bolster sustainability efforts within this industry.

Additionally, the chapter emphasizes potential areas for future investigation, inviting fellow researchers to delve deeper into specific facets that surfaced during this study. Ultimately, this final chapter acts as a synthesis of insights, presenting conclusions, actionable advice, and avenues for further exploration within the realm of sustainable development in bio-based industries.

7.1 Conclusions of the study

7.1.1 Impact & relationship between Policy Frameworks and Implementation and Sustainable Development of Bio-based Industries

In this study, Policy Frameworks and Implementation were found to have a moderately positive impact on the Sustainable Development of Bio-based Industries in Sri Lanka. This means that these policies play a role in positively influencing the industry's sustainable growth, explaining about 9.4 % of the potential impact according to the analysis conducted in chapter five. Examining various aspects like Regulatory Guidance, Incentives and Support, Stakeholder Engagement, Long-Term Planning, Global Competitiveness, and Addressing Challenges, it's evident that Policy Frameworks and Implementation contribute positively to shaping the industry's sustainable practices.(Dorlach & Gunasekara, 2023)

Analyzing respondents' perceptions through the questionnaire revealed a consensus among them regarding Policy Frameworks and Implementation. Their responses consistently leaned towards agreement, indicating a shared perception with minimal variation among their answers. While Policy Frameworks and Implementation do make a positive contribution to the development of Sustainable Bio-based Industries in Sri Lanka, it's noteworthy that compared to the other factors explored in the study, their impact seems relatively lower. This suggests that while these policies are influential, other factors might have a more significant role in driving sustainable growth within the bio-based industry in Sri Lanka.(Timonina & Stephan, 2023; Wanigasundera & Atapattu, 2019)

7.1.2 Impact & relationship between Technological Advancements and Innovations and Sustainable Development of Bio-based Industries

The study discovered a significantly strong and positive connection between Technological Advancements and Innovations and the Sustainable Development of Bio-based Industries in Sri Lanka. This indicates that advancements in technology play a pivotal role in driving the industry's sustainability, accounting for a substantial 70.2 % chance to impact the industry's sustainable development, as revealed in the fourth chapter's analysis. The various facets of Technological Advancements and Innovations, including Research and Development, New Materials and Processes, Automation and Robotics, Biorefineries, Sustainable Practices, Biotechnology Applications, Digitalization and Data Analytics, Collaborative Platforms, and Adoption of Green Technologies, collectively contribute to this impactful relationship.

The respondents' consensus, as reflected in their questionnaire responses, consistently indicated an agreement regarding the significance of technological advancements. Their perceptions aligned closely, evident from the minimal standard deviation values, suggesting a shared understanding among stakeholders, policymakers, and industry representatives. This study was targeted towards policymakers, bio-based industry stakeholders, and local farmers, and the results overwhelmingly confirm the strongly positive relationship between technological advancements and sustainable development. As a result, it's evident that prioritizing investments and efforts in Technological Advancements and Innovations is crucial for the success and effectiveness of Sustainable Bio-based Industry activities in Sri Lanka. This factor emerges as the most impactful variable, urging a higher emphasis on technological innovation for the industry's sustainable development. (Asghar et al., 2022; Escobar & Laibach, 2021)

7.1.3 Impact & relationship between Stakeholder Engagement and Participation and Sustainable Development of Bio-based Industries

The study uncovered a significant and moderately positive relationship between Stakeholder Engagement and Participation and the Sustainable Development of Bio-based Industries in Sri Lanka. This indicates that actively involving stakeholders indeed impacts the industry's sustainability, accounting for about 48.4 % chance to influence the industry's sustainable development, as evidenced in the analysis conducted in chapter five. The facets within Stakeholder Engagement and Participation, including Community Involvement, Industry Collaboration, Government and Regulatory Bodies, Supply Chain Collaboration, Public Awareness and Education, NGOs and Advocacy Groups, Research and Academic Partnerships, Transparent Communication, Ethical Considerations, and Continuous Improvement, collectively contribute to this impactful relationship.

The questionnaire responses regarding Stakeholder Engagement and Participation showed a consensus among respondents, consistently indicating agreement. Their shared perception, reflected in the minimal standard deviation values, signifies a common understanding among stakeholders, policymakers, and industry

players. Given these findings, it's apparent that Sri Lanka stands to advance its Sustainable Development of Bio-based Industries by placing significant emphasis on Stakeholder Engagement and Participation. Actively involving diverse stakeholders across various aspects of the industry can drive positive changes, foster sustainability, and ensure a more inclusive and responsible approach to bio-based industry practices.(Mandolesi et al., 2023)

7.1.4 Impact & relationship between Stakeholder Engagement and Participation and Sustainable Development of Bio-based Industries

The study reveals a significant and moderately positive relationship between Market Opportunities and Access and the Sustainable Development of Bio-based Industries in Sri Lanka. This signifies that Market Opportunities and Access hold a notable influence on advancing the industry's sustainability, accounting for approximately 25.5 % chance to impact the industry's sustainable development, as analyzed in chapter four.

The components within Market Opportunities and Access, such as Market Research and Analysis, Market Segmentation, Value Chain Integration, Supply and Demand Dynamics, Market Entry Strategies, Distribution Channels, Consumer Education and Marketing, International Trade and Export Opportunities, Partnerships and Alliances, and Adaptability and Innovation, collectively contribute to this impactful relationship.

Analysis of responses from the Market Opportunities and Access-related questionnaire revealed a consensus among respondents, indicating a prevailing agreement. Their shared perception, as indicated by the minimal standard deviation values, suggests that employees are content with the organization's Market Opportunities and Access strategies aimed at fostering the Sustainable Development of Bio-based Industries in Sri Lanka. This alignment among employees signifies satisfaction with the strategies implemented to tap into market opportunities and ensure access, setting a positive direction for achieving sustainable development within the bio-based industry in Sri Lanka.(Dammer et al., 2013)

7.1.5 Summary of the Conclusions

Across the facets studied, each plays a distinct role in shaping sustainable development within Sri Lanka's bio-based industries. Policy Frameworks and Implementation, while showcasing a positive influence on sustainability, contribute about 9.4 % to its potential impact. Yet, compared to other elements explored, its effect appears relatively moderate, hinting at the presence of more influential factors in the industry's sustainable growth.

Conversely, Technological Advancements and Innovations stand out significantly, demonstrating a robust and powerful relationship, attributing a substantial 70.2 % chance to impact sustainable growth. This underscores the critical need for substantial investments and focused efforts in technological innovation to drive the industry's success in sustainable development.

Stakeholder Engagement and Participation also emerge as vital contributors, showcasing a significant and moderately positive impact, accounting for about 48.4 % chance to influence sustainability. This emphasizes the pivotal role of actively involving a diverse range of stakeholders across various dimensions, fostering a more inclusive and responsible approach within the bio-based industry.

Furthermore, Market Opportunities and Access play a noteworthy role, accounting for approximately 25.5 % chance to impact sustainable development. The strategies employed to access markets and leverage opportunities receive positive feedback, reflecting employee satisfaction and setting a favorable course toward achieving sustainability within the industry.

In essence, while technological advancements surface as the most impactful factor, a comprehensive approach that integrates policy frameworks, stakeholder engagement, and market access is pivotal for driving holistic and sustainable growth within Sri Lanka's bio-based industries. Each element contributes uniquely, highlighting the necessity of a well-rounded strategy to propel the industry's sustainable development forward.

7.2 Recommendations

Based on the in-depth analysis conducted in this study, several key recommendations emerge to advance the Sustainable Development of Bio-based Industries in Sri Lanka. By considering the Strategic Investment in Technology, Recognizing the significant impact of Technological Advancements and Innovations on sustainable growth, it is crucial to channel more resources into research and development areas such as biotechnology, automation, sustainable practices, and digitalization within the industry. By encouraging innovation in these domains, Sri Lanka can effectively harness technological advancements to drive sustainable practices within the bio-based sector.

By considering the Enhanced Collaboration Among Stakeholders, strengthening partnerships and collaborations among a diverse range of stakeholders, including government bodies, industry players, local communities, NGOs, academia, and advocacy groups, is pivotal for sustainable growth. Increasing initiatives that involve these stakeholders foster a more inclusive and responsible industry approach, ensuring that bio-based practices align with sustainability goals and benefit various stakeholders.

The topic of Strategic Planning for Market Access: Acknowledging the significant impact of Market Opportunities and Access, a strategic approach to planning becomes imperative. This involves conducting thorough market research, implementing effective market segmentation, and fortifying distribution channels. Additionally, fostering consumer education initiatives is essential to maximize the industry's impact by effectively tapping into emerging market trends and opportunities.

According to the Refinement of Policy Frameworks, While Policy Frameworks and Implementation exhibit a moderately positive impact, there's a need for

refinement. Strengthening existing policies by providing clearer regulatory guidance, enhancing incentives, and incorporating long-term planning aligned with global competitiveness can substantially enhance their influence on fostering sustainable practices within the bio-based industry.

As per the Cultivating a Culture of Continuous Improvement, encouraging a culture of ongoing improvement within bio-based industries is essential for long-term sustainability. This involves prioritizing transparency, ethical considerations, and adaptive strategies that align with evolving market and technological trends. By embracing a continuous improvement approach, the industry can ensure sustained progress and resilience against challenges, fostering an environment conducive to long-term, sustainable growth.

By prioritizing and implementing these recommendations, Sri Lanka's bio-based industries can establish a robust foundation for sustainable growth. These concerted efforts not only improve competitiveness but also contribute to creating an environmentally conscious and inclusive industry landscape, making a substantial impact on the nation's sustainable development objectives.

7.3 Open Avenue for other research

The findings of this study open doors to several promising research directions that can expand our understanding of Sustainable Development within Sri Lanka's Bio-based Industries. These potential paths cover various key areas.

Firstly, a detailed Policy Analysis offers a chance to enhance Sri Lanka's bio-based industry policies. This involves closely examining specific policy mechanisms, their evolution, and comparing them with global policies to strengthen strategies for sustainable development. Secondly, a focus on Technological Impact Assessment can map out technological contributions and predict the influence of emerging technologies. Comparative studies on global technological advancements can offer valuable insights. Thirdly, deeper research into Stakeholder Dynamics and Engagement can unveil effective participation strategies. Understanding how stakeholders' perceptions influence sustainability can guide more efficient engagement models. Furthermore, investigating Market Access and Innovation Strategies can reveal insights into consumer behavior and innovative market approaches. Comparative studies with global industries could provide valuable benchmarks. Additionally, exploring the correlation between Employee Satisfaction and Organizational Strategies offers insight into how market strategies influence sustainable development and organizational culture. Lastly, embracing Interdisciplinary Approaches by merging technology, policy, social sciences, and environmental studies can offer comprehensive insights and frameworks for sustainable bio-based industry development. These research avenues, pursued by scholars and industry experts, promise to enrich our knowledge of sustainability in bio-based industries. They hold potential to foster innovation, refine policies, enhance stakeholder engagement, and understand market dynamics in Sri Lanka. Ultimately, this collective knowledge will propel the country towards more sustainable practices in its bio-based industries.

8 References

- Abeywardana, N., Schütt, B., Wagalawatta, T., & Bebermeier, W. (2019). Indigenous agricultural systems in the dry zone of Sri Lanka: Management transformation assessment and sustainability. *Sustainability*, *11*(3), 910.
- Amaratunga, D., Fernando, N., Haigh, R., & Jayasinghe, N. (2020). The COVID-19 outbreak in Sri Lanka: A synoptic analysis focusing on trends, impacts, risks and science-policy interaction processes. *Progress in Disaster Science*, *8*, 100133.
- Asghar, A., Sairash, S., Hussain, N., Baqar, Z., Sumrin, A., & Bilal, M. (2022). Current challenges of biomass refinery and prospects of emerging technologies for sustainable bioproducts and bioeconomy. *Biofuels, Bioproducts and Biorefining*, *16*(6), 1478–1494. <https://doi.org/10.1002/bbb.2403>
- Burchfield, E. K., & Poterie, A. T. de la. (2018). Determinants of crop diversification in rice-dominated Sri Lankan agricultural systems. *Journal of Rural Studies*, *61*, 206–215. <https://doi.org/10.1016/j.jrurstud.2018.05.010>
- Cairney, P., Timonina, I., & Stephan, H. (2023). How can policy and policymaking foster climate justice? A qualitative systematic review. *Open Research Europe*, *3*, 51. <https://doi.org/10.12688/openreseurope.15719.2>
- D’Amico, G., Szopik-Depczyńska, K., Beltramo, R., D’Adamo, I., & Ioppolo, G. (2022). Smart and Sustainable Bioeconomy Platform: A New Approach towards Sustainability. *Sustainability*, *14*(1), Article 1. <https://doi.org/10.3390/su14010466>

- Dammer, L., Carus, M., Raschka, A., & Scholz, L. (2013). *Market Developments of and Opportunities for biobased products and chemicals*.
- Dietz, T., Börner, J., Förster, J. J., & Von Braun, J. (2018). Governance of the bioeconomy: A global comparative study of national bioeconomy strategies. *Sustainability*, 10(9), 3190.
- Dissanayake, D., Tilt, C. A., & Qian, W. (2021). How do public companies respond to national challenges through sustainability reporting? – The case of Sri Lanka. *Qualitative Research in Accounting & Management*, 18(4/5), 455–483.
<https://doi.org/10.1108/QRAM-06-2020-0088>
- Dorlach, T., & Gunasekara, S. (2023). The politics of glyphosate regulation: Lessons from Sri Lanka's short-lived ban. *Globalization and Health*, 19(1), 84.
<https://doi.org/10.1186/s12992-023-00981-2>
- Drechsel, P., Madhuwanthi, P., Nisansala, D., Ramamoorthi, D., & Bandara, T. (2023). *On the feasibility of an agricultural revolution: Sri Lanka's move to go 100% organic*.
- English, C. (2021, September 9). *All-Organic Agriculture? Sri Lanka Cripples Farmers and Sparks Food Shortage*. American Council on Science and Health.
<https://www.acsh.org/news/2021/09/09/all-organic-agriculture-sri-lanka-cripples-farmers-and-sparks-food-shortage-15796>
- Eriyagama, N., Smakhtin, V., Chandrapala, L., & Fernando, K. (2010). *Impacts of climate change on water resources and agriculture in Sri Lanka: A review and preliminary vulnerability mapping*.

- Escobar, N., & Laibach, N. (2021). Sustainability check for bio-based technologies: A review of process-based and life cycle approaches. *Renewable and Sustainable Energy Reviews*, 135, 110213. <https://doi.org/10.1016/j.rser.2020.110213>
- Esham, M., Jacobs, B., Rosairo, H. S. R., & Siddighi, B. B. (2018). Climate change and food security: A Sri Lankan perspective. *Environment, Development and Sustainability*, 20, 1017–1036.
- Ferrando, T., Jokubauskaite, G., Rossati, D., & De Feyter, K. (2022). *The Belgian Investment Company for Developing Countries (BIO) as a sustainable development actor*.
- Filho, W., & Brandli, L. (2016). *Engaging Stakeholders for Sustainable Development*. https://doi.org/10.1007/978-3-319-26734-0_21
- Fobbe, L., & Hilletofth, P. (2021). The role of stakeholder interaction in sustainable business models. A systematic literature review. *Journal of Cleaner Production*, 327, 129510. <https://doi.org/10.1016/j.jclepro.2021.129510>
- Gamage, A., Gangahagedara, R., Gamage, J., Jayasinghe, N., Kodikara, N., Suraweera, P., & Merah, O. (2023). Role of organic farming for achieving sustainability in agriculture. *Farming System*, 1(1), 100005. <https://doi.org/10.1016/j.far-sys.2023.100005>
- Gunarathne, D. S., Udugama, I. A., Jayawardena, S., Gernaey, K. V., Mansouri, S. S., & Narayana, M. (2019). Resource recovery from bio-based production processes in developing Asia. *Sustainable Production and Consumption*, 17, 196–214. <https://doi.org/10.1016/j.spc.2018.11.008>

- Kurian, M., & Kojima, Y. (2021). Chapter 2—Blind spots in environmental governance. In M. Kurian & Y. Kojima (Eds.), *Boundary Science* (pp. 27–74). Elsevier. <https://doi.org/10.1016/B978-0-323-88473-0.00002-6>
- Leipold, S., & Petit-Boix, A. (2018). The circular economy and the bio-based sector—Perspectives of European and German stakeholders. *Journal of Cleaner Production*, 201, 1125–1137. <https://doi.org/10.1016/j.jclepro.2018.08.019>
- Lemke, A. A., & Harris-Wai, J. N. (2015). Stakeholder engagement in policy development: Challenges and opportunities for human genomics. *Genetics in Medicine : Official Journal of the American College of Medical Genetics*, 17(12), 949–957. <https://doi.org/10.1038/gim.2015.8>
- Mandolesi, S., Kilic, B., Naspetti, S., & Zanolli, R. (2023). Switching to bio-based packaging for organic products: Supply chain actors' perspectives. *Organic Agriculture*. <https://doi.org/10.1007/s13165-023-00451-4>
- Melvani, K., Bristow, M., Moles, J., Crase, B., & Kaestli, M. (2020). Multiple livelihood strategies and high floristic diversity increase the adaptive capacity and resilience of Sri Lankan farming enterprises. *Science of the Total Environment*, 739, 139120.
- Moosmann, D., Majer, S., Ugarte, S., Ladu, L., Wurster, S., & Thrän, D. (2020). Strengths and gaps of the EU frameworks for the sustainability assessment of bio-based products and bioenergy. *Energy, Sustainability and Society*, 10(1), 22. <https://doi.org/10.1186/s13705-020-00251-8>
- Munasinghe, M., Deraniyagala, Y., Dassanayake, N., & Karunarathna, H. (2017). Economic, social and environmental impacts and overall sustainability of the tea

- sector in Sri Lanka. *Sustainable Production and Consumption*, 12, 155–169.
<https://doi.org/10.1016/j.spc.2017.07.003>
- Nagothu, U. S. (2020). *The Bioeconomy Approach: Constraints and Opportunities for Sustainable Development*. Routledge.
- Paddy Statistics. 2021 Yala Season. Department of Census and Statistics Sri Lanka.
 (n.d.). <http://www.statistics.gov.lk/Resource/en/Agriculture/paddystatistics/PaddyStatsPages/2021yala.pdf>
- Pandey, P., & Sharma, A. (2021). Knowledge politics, vulnerability and recognition-based justice: Public participation in renewable energy transitions in India. *Energy Research & Social Science*, 71, 101824.
- Panneerselvam, P., Halberg, N., & Lockie, S. (2012). Consequences of organic agriculture for smallholder farmers' livelihood and food security. In *Organic agriculture for sustainable livelihoods* (pp. 21–44). Routledge.
- Rafiaani, P., Kuppens, T., Van Dael, M., Azadi, H., Lebailly, P., & Van Passel, S. (2018). Social sustainability assessments in the biobased economy: Towards a systemic approach. *Renewable and Sustainable Energy Reviews*, 82, 1839–1853.
- Råheim, M., Magnussen, L. H., Sekse, R. J. T., Lunde, Å., Jacobsen, T., & Blystad, A. (2016). Researcher–researched relationship in qualitative research: Shifts in positions and researcher vulnerability. *International Journal of Qualitative Studies on Health and Well-Being*, 11(1), 30996. <https://doi.org/10.3402/qhw.v11.30996>
- Ranathilaka, M. B., & Arachchi, I. A. J. I. (2019). THE EFFECT OF FERTILIZER SUBSIDY ON PADDY PRODUCTION OF SMALL SCALE FARMERS: SPECIAL REFERENCE IN POLONNARUWA DISTRICTIN SRI LANKA.

Review of Behavioral Aspect in Organizations and Society, 1(1), Article 1.

<https://doi.org/10.32770/rbaos.vol133-44>

Rathnachandra, S., Sivashankar, P., & Malkanthi, S. (2023). Circular Economy for Sustainable Agricultural Development in Sri Lanka. *Multisectoral Approaches to Accelerate Economic Transformation in the Face of Crisis in Sri Lanka*, 176.

Rathnayake, S., Gray, D., Reid, J., & Ramilan, T. (2022). The impacts of the COVID-19 shock on sustainability and farmer livelihoods in Sri Lanka. *Current Research in Environmental Sustainability*, 4, 100131. <https://doi.org/10.1016/j.cr-sust.2022.100131>

Rodrigo, M. (2022, July 19). *Sri Lankan environmental policy failures helped fuel people power revolution*. Mongabay Environmental News. <https://news.mongabay.com/2022/07/in-sri-lanka-environmental-policy-failures-helped-fuel-a-people-power-revolution/>

Rosairo, H. (2023). Smallholder Agriculture in Developing and Emerging Economies: The Case of Sri Lanka. In *Sustainable Food Value Chain Development: Perspectives from Developing and Emerging Economies* (pp. 259–293). Springer.

Roshan, A. S. M., & Pandithage, N. W. S. K. (2023). *Good Agricultural Practices in Sri Lanka: Status, Challenges and Way Forward*.

Sample Size Calculator—Confidence Level, Confidence Interval, Sample Size, Population Size, Relevant Population—Creative Research Systems. (n.d.). Retrieved December 31, 2023, from <https://www.surveysystem.com/sscalc.htm>

- Senanayake, S. M. P., & Premaratne, S. P. (2016). An Analysis of the Paddy/Rice Value Chains in Sri Lanka. *Asia-Pacific Journal of Rural Development*, 26(1), 105–126. <https://doi.org/10.1177/1018529120160104>
- Singh, P. (2022, September 12). *Food insecurity in Sri Lanka likely to worsen amid poor agricultural production, price spikes and ongoing economic crisis, FAO and WFP warn*. Newsroom. <https://www.fao.org/newsroom/detail/food-insecurity-in-sri-lanka-likely-to-worsen-amid-poor-agricultural-production-price-spikes-and-ongoing-economic-crisis-fao-and-wfp-warn/en>
- Sri Lanka: Grain and Feed Annual* | USDA Foreign Agricultural Service. (2022, September 1). <https://fas.usda.gov/data/sri-lanka-grain-and-feed-annual-2>
- Timonina, I., & Stephan, H. (2023, October 10). *How can policy and policymaking foster ...* | *Open Research Europe*. <https://open-research-europe.ec.europa.eu/articles/3-51/v2>
- Upadhyay, N., Tripathi, S., Kushwaha, A., Bhasney, S. M., & Mishra, M. (2023). Chapter 2 - Renewable bio-based materials: A journey towards the development of sustainable ecosystem. In C. M. Hussain, A. Kushwaha, R. N. Bharagava, & L. Goswami (Eds.), *Bio-Based Materials and Waste for Energy Generation and Resource Management* (Vol. 5, pp. 31–75). Elsevier. <https://doi.org/10.1016/B978-0-323-91149-8.00005-3>
- Vinci, G., Ruggieri, R., Ruggeri, M., & Prencipe, S. A. (2023). Rice Production Chain: Environmental and Social Impact Assessment—A Review. *Agriculture*, 13(2), 340.

- Wanigasundera, W., & Atapattu, N. (2019). Extension reforms in Sri Lanka: Lessons and policy options. In *Agricultural extension reforms in south asia* (pp. 79–98). Elsevier.
- Weerahewa, J., & Dayananda, D. (2023). Land use changes and economic effects of alternative fertilizer policies: A simulation analysis with a bio-economic model for a Tank Village of Sri Lanka. *Agricultural Systems*, 205, 103563. <https://doi.org/10.1016/j.agry.2022.103563>
- Wijerathna-Yapa, A., Henry, R. J., Dunn, M., & Beveridge, C. A. (n.d.). Science and opinion in decision making: A case study of the food security collapse in Sri Lanka. *Modern Agriculture*, n/a(n/a). <https://doi.org/10.1002/moda.18>
- Wijetunga, C. S., & Saito, K. (n.d.). *Evaluating the Fertilizer Subsidy Reforms in the Rice Production Sector in Sri Lanka: A Simulation Analysis*.
- Williams, N. E., Dhamruwan, M., & Carrico, A. R. (2023). Displacement and degradation: Impediments to agricultural livelihoods among ethnic minority farmers in post-war Sri Lanka. *Ambio*, 52(4), 813–825. <https://doi.org/10.1007/s13280-022-01819-8>

9 List of figures

Fig. 1 Total production of rice harvest per year	13
Fig. 2 No. of complaints to the Ministry of agriculture regarding the bio-based farming issues per year Source: (<i>Sri Lanka</i> , 2022)	14
Fig. 3 Raw rice production Vs imported quantity	15
Fig. 4 Conceptual Framework of the study.....	26
Fig. 5 Conceptual Framework of the study	26
Fig. 6 Sample Calculator.....	30
Fig. 7 Population to samples.....	32
Fig. 8 Population Structure and Age Distribution.....	37
Fig. 9 Region wise percentages of the respondents.....	38
Fig. 10 Gender Dimensions.....	39
Fig. 11 Frequency distribution, Histogram & normality curve of the EMMS data	41
Fig. 12 Regression analysis of the policy frameworks and Implementation on SDBI	43
Fig. 13 Regression analysis of the Technological Advancements and Innovations on SDBI	46
Fig. 14 Regression analysis of the Stakeholder Engagement and participation on SDBI	48
Fig. 15 Regression analysis of the Market Opportunities and Access on SDBI.....	51
Fig. 16 Multiple Regression analysis.....	53

10 List of tables

Tab. 1 Total production of rice harvest per year	13
Tab. 2 No. of complaints to the Ministry of agriculture regarding the bio-based farming issues per year	14
Tab. 3 Raw rice production Vs imported quantity	15
Tab. 4 Hypothesis as per the conceptual framework of the study	27
Tab. 5 Operationalization of the study	28
Tab. 6 Sample calculation.....	32
Tab. 7 Data collection summary	34
Tab. 8 Results & status of the reliability test	36
Tab. 9 Population Structure and Age Distribution	37
Tab. 10 Population Structure and Age Distribution.....	38
Tab. 11 Gender Dimensions	39
Tab. 12 Grand Mean, Standard deviation & Skewness of the S D B I.....	40
Tab. 13 Correlation analysis of the policy frameworks and Implementation on SDBI	42
Tab. 14 Regression analysis of the policy frameworks and Implementation on SDBI	43
Tab. 15 ANOVA of the policy frameworks and Implementation on SDBI.....	44
Tab. 16 Correlation analysis of the Technological Advancements and Innovations on SDBI	44
Tab. 17 Regression analysis of the Technological Advancements and Innovations on SDBI	45
Tab. 18 ANOVA of the Technological Advancements and Innovations on SDBI.....	46
Tab. 19 Correlation analysis of the Stakeholder Engagement and Participation on SDBI	47
Tab. 20 Regression analysis of the Stakeholder Engagement and Participation on SDBI	48
Tab. 21 ANOVA of the Stakeholder Engagement and Participation on SDBI	49
Tab. 22 Correlation analysis of the Market Opportunities and Access on SDBI	49
Tab. 23 Regression analysis of the Market Opportunities and Access on SDBI	50

Tab. 24 ANOVA of the Market Opportunities and Access on SDBI.....	51
Tab. 25 Multiple Regression analysis	52
Tab. 26 Multiple ANOVA analysis	52
Tab. 27 Multiple results of the coefficients & significant values.....	52
Tab. 28 Summary of the hypothesis test, Source: SPSS.....	54

Attachments

Questionnaire Form (blank)

The Bio-blunder and Policy Failure: A case study of Sri Lanka

Dear member,

I am a university student and have to analysis of the factors determining the Sustainable Development of Bio-based Industries.

You are invited to participate this in research. In the study, you will be asked to complete a questionnaire. I would appreciate it if you can spare few minutes and fill this questionnaire which is used to gather data for my research.

I guarantee that the information you provide will not be any third party and will only be used for the purpose of my research. I appreciate very much your corporation in this regard.

1.0 General Information

Please tick (✓) the box to confirm your answer.

1.1 Your Age:

18 yrs - 25 yrs	25 yrs - 35 yrs	35 yrs - 45 yrs	45 yrs - 55 yrs	More than 55 yrs

1.2 Your Working Experience:

Below 1 year	01 yr - 05 yrs	05 yrs - 10 yrs	10 yrs - 15 yrs	15 yrs - 20 yrs	More than 20 yrs

1.3 You are working in

Rural Area	Sub urban Area	Urban Area

1.4 Your Gender:

Male	Female

1.5 Your Education Background:

School Level	Certificate Level	Diploma level	Graduated level	Beyond the Graduated level

1.6 What are your thoughts& ideas about the Sustainable Development of Bio-based Industries?

.....

.....

.....

2.0 Sustainable Development of Bio-based Industries

Q : Statement	HD	D	N	A	HA
2.1: How would you rate the effectiveness of current practices in promoting environmental sustainability within bio-based industries?					
2.2: Evaluate the economic viability of bio-based industries.					
2.3: To what extent do you perceive the economic viability contributing to the overall sustainability of bio-based industries?					
2.4: Assess the social impact of bio-based industries					
2.5: Rate the contribution of bio-based industries to social development and well-being.					

Where, **HD** -Highly Disagreed (1), **D** - Disagreed (2), **N** - Neutral (3), **A** - Agreed (4), **HA** - Highly Agreed (5)

2.6: What is your idea of the level Sustainable Development of Bio-based Industries?

.....

.....

.....

3.0 Identified factors.

3.1 Policy frameworks and implementation

Q : Statement	HD	D	N	A	HA
3.2.1: To what extent do you perceive the coherence of existing policy frameworks related to bio-based industries?					
3.2.2: How effectively do you think these policies are implemented?					
3.2.3: Rate the effectiveness of current policies in fostering sustainable development in bio-based industries.					
3.2.4: Evaluate the impact of policy measures on the overall sustainability of bio-based industries.					
3.2.5: On a scale of 1 to 5, assess the level of compliance and enforcement of policies governing bio-based industries.					

Where, **HD** -Highly Disagreed (1), **D** - Disagreed (2), **N** - Neutral (3), **A** - Agreed (4), **HA** - Highly Agreed (5)

3.1.6: What is your suggestion to Sustainable Development of Bio-based Industries effective level through the Policy frameworks and implementation?

.....

.....

.....

3.2 Technological advancements and innovations

Q : Statement	HD	D	N	A	HA
3.2.1: Rate the level of research and development investment in bio-based industries					
3.2.2: How would you rate the current level of patent filings and intellectual property development in this sector?					
3.2.3: Assess the rate of technology adoption in bio-based industries.					
3.2.4: To what extent do you perceive the effectiveness of technology adoption in enhancing sustainability?					
3.2.5: Rate the impact of technological innovations on the overall development of bio-based industries					

Where, **HD** –Highly Disagreed (1), **D** – Disagreed (2), **N** – Neutral (3), **A** – Agreed (4), **HA** - Highly Agreed (5)

3.2.6: What is your suggestion to Sustainable Development of Bio-based Industries effective level through the Technological advancements and innovations?

.....

.....

.....

3.3 Stakeholder engagement and participation

Q : Statement	HD	D	N	A	HA
3.3.1: Evaluate the representation of stakeholders in the decision-making processes of bio-based industries.					
3.3.2: On a scale of 1 to 5, assess the level of public awareness and engagement in bio-based industries					
3.3.3: Rate the effectiveness of collaboration and partnerships within the bio-based industries.					
3.3.4: To what extent do you perceive partnerships contributing to sustainable development in this sector?					
3.3.5: Your rate of the quality of the stakeholders relevant to the topic.					

Where, **HD** –Highly Disagreed (1), **D** – Disagreed (2), **N** – Neutral (3), **A** – Agreed (4), **HA** - Highly Agreed (5)

3.3.6: What is your suggestion to Sustainable Development of Bio-based Industries effective level through the Stakeholder engagement and participation?

.....

.....

.....

.....

3.4 Market opportunities and access

Q : Statement	HD	D	N	A	HA
3.4.1: Assess the growth rate of bio-based industries in the current market scenario					
3.4.2: On a scale of 1 to 5, evaluate the intensity of market competition in bio-based industries.					
3.4.3: Rate the level of market access barriers faced by bio-based industries.					
3.4.4: To what extent do these barriers impact the sustainable development of bio-based industries?					
3.4.5: Market opportunities and access is relating to the bio-based industries development					

Where, **HD** –Highly Disagreed (1), **D** – Disagreed (2), **N** – Neutral (3), **A** – Agreed (4), **HA** - Highly Agreed (5)

3.4.6: What is your idea of the Market opportunities and access of the bio industries?

.....

.....

.....

.....

*** Thank You ***